

Neural Rendering Composition and Granularity Tuning: State- and Task-Dependent Assembly of Multimodal Memory and Experience

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Abstract

A neural rendering, as proposed in Self-Aware Networks (SAN), is an evolving assembly of perceptual, remembered, bodily, conceptual, emotional and prospective relations, not merely a visual image. I propose Neural Rendering Composition and Granularity Tuning to distinguish which contents participate, how they are bound, which differences survive and how far an episode extends in time. The candidate contribution is a specific decomposition and measurement program, not the discovery of task-dependent cognition. Seven elementary propositions and a bounded twenty-statement Lean supplement clarify sufficiency, refinement, incomparability, observation and future-query requirements; restricted readout failure need not imply lost information. Twelve trained recurrent networks demonstrate differences among primary performance, broader access and cross-role reuse, but the matched-context models remain at chance on the intended temporal relation. A development-only diagnosis does not rescue that failure. A subsequent conventional active classifier learns coarse/fine queries and responses while retaining only encountered evidence. Across three fitted seeds and separate background sets, its second-query accuracy ties a fine-only learner at 0.857 while acquiring 1.961 versus 2.500 nominal new source bits. A complete-record control performs better, at 1.000, with greater acquisition cost. The relation is answered correctly when the needed earlier observation was retained and remains at chance when it was not. Assembly-only interventions alter responses without changing the source record, while equivalent attention and inverse-restoration controls tie exactly. These symbolic results make the proposed distinctions inspectable; they neither repair the recurrent implementation nor establish a new neural algorithm. The biological hypothesis concerns learned reception, temporary participation, relational organization, active sampling and returned consequences jointly configuring an episode. Published-method comparisons, independently collected measurements and causal evidence remain necessary. Mathematical consistency, conventional learned sampling and synthetic accuracy do not establish conscious experience.

1 The control problem is larger than a sharper image

The same basketball arena can support very different experienced and actionable situations. A player may reconstruct body position, the ball's trajectory, a teammate's anticipated movement, and the opening for a pass. A security worker may instead maintain relationships among entrances, unusual movements, responsibilities, and response routes. An event manager must coordinate schedules, locations, people, and

unresolved dependencies. A speaker must connect a developing argument with remembered examples, vocal production, and audience response. These examples motivate the proposal; they are not observations collected for this study.

What changes is not adequately described as the sharpness of a visual scene. Some details become available, others cease to matter, and different relationships organize the same setting. A narrow episode can be elaborate around one object while omitting much of its surroundings. A broad episode can span many actors while representing each with modest detail. Temporal depth can change independently: an immediate movement and a remembered sequence may recruit overlapping content but organize it differently.

In SAN, neural rendering already includes nonvisual and internally generated content [1-5]. The present proposal does not add multimodality to a previously visual theory. It makes an existing control problem explicit: how much of a learned repertoire becomes effective now, at what granularity, in which relationships, over what interval, and with what consequence? The title adds *composition* because the question concerns the organization of an episode, not only its smallest detectable difference. The word *and* avoids reducing composition and granularity to one variable.

This is a system-level hypothesis with a deliberately limited first implementation. Its value must ultimately depend on discriminating evidence, not the breadth of the situations it can describe. Section 5 reports a conventional-attention tie that prevents the present application from establishing algorithmic novelty.

2 Provenance and the closest existing explanations

2.1 The SAN lineage and its limits

An internally dated September 2011 dialogue describes a sensory, statistical-memory, brainwave, action, and returned-sensation loop and calls the self a learning loop that learns itself. An internally dated September 2012 note develops nested learning loops. A further composite note, associated with the Telescoping Mind discussion, considers different components contributing at different scales and abstraction levels. These are partial conceptual ancestors, not records of the present named framework [6-8]. The current files contain later additions. Their internal dates do not date every sentence, and this draft has not independently re-established their earliest public fixation.

A more direct ancestor is my note a0258z in a Git commit dated July 5, 2022 [19]. Alongside associations, recalled patterns and prospective reconstruction, it proposes that successive excitatory and inhibitory phase events refine a modeled pattern. It explicitly discusses a granularity limitation through a Taylor-series analogy. This is an early proposal about evolving representational detail, not a proof of a neuronal Taylor algorithm or the present composition-and-granularity model. The exact historical bytes and recorded commit date have been checked; current public access to them is not independent proof of their first public availability.

Five existing papers create substantial overlap obligations. Telescoping Network Reinstatement develops local-to-larger reinstatement with dynamic recruitment and recurrent reconstruction [1]. Phase-Wave Differential Calculus specifies receiver-relative events, tuning-state updates and local approximation [2]. The Qualia Tomogram addresses receiver-conditioned reconstruction [3]. Fractal Conscious Perception already develops multiscale, multimodal relations, including the broccoli example [4]. The Learning Loop That Learns Itself joins self-world reinstatement to embodied feedback and learning [5]. None should be redescribed as having ignored multimodality or change over time merely to create novelty here.

In the selected public version of Telescoping Network Reinstatement, equation 10 uses time-varying recruitment weights and equation 12 updates a tonic canvas using phasic contributions, learned state, inhibition, action and outcome [1]. Its section 10.6 also asks how an unchanged cue produces different reconstructions under different goals and bodily conditions. A newer local revision retains those operations; they are not new merely because this paper groups some of their consequences into configuration coordinates. Phase-Wave Differential Calculus already separates immediate state dependence, temporary effective routing, and persistent plastic change [2]. Those distinctions constrain, rather than originate in, the present proposal.

Full-text review of Fractal Conscious Perception makes this overlap more specific. Its cross-scale account already uses task-weighted relation preservation, learned receiving conditions, and selective reconstruction; its equations 9-10 cannot be summarized as a static, task-independent scaling story [4]. Its separate focus is the added value of typed finite-range scaling features and route-specific evidence. Task relevance and multimodal participation themselves are therefore not new contributions here.

The Qualia Tomogram creates a particularly substantial overlap obligation [3]. Its sections 29-31 already join learned receiver conditions, temporary routes, typed PWD contributions, multimodal content, boundaries, memory reinstatement and active sampling. Its content tuple includes bodily state, remembered relations and possible actions; its conscious-entry discussion separates available memory from current recruitment. It also distinguishes several temporal endpoints and supplies mathematical ambiguity and coordinate-change arguments. Consequently, neither an assembled multimodal episode nor the separation of available memory from present expression originates in this paper. Its synthetic inverse is close to a raw linear comparator, which is slightly better on average across six smaller replications; an adverse generator favors amplitude geometry. Those boundaries remain relevant here.

The Learning Loop That Learns Itself already includes a learned application, not just a verbal feedback cycle [5]. Its second synthetic study learns an arbitrary cue/action relation after a shortcut was identified in the first study. It tests persistent learning, partial cues, perturbation and rescue, while its matched generic recurrent learner ties the full model exactly. Another learned value table, a closed loop or an invertible rescue would therefore not establish a new contribution by itself. The present application specifically separates acquired associations, stored observations, temporary composition/granularity and a downstream readout, while preserving the possibility that an established attention account implements the same operations.

The proposed residual is a *specific decomposition and test of episodic reconfiguration*: when an organism changes role or state, which contents and bindings become effective, which distinctions survive, how far an episode extends in time, and whether these changes can be distinguished from a different readout of unchanged available information. Time variation, task dependence and recurrent reconstruction are inherited parts of the question. Full extracted-text review is complete for all five selected public manuscripts and the newer local Paper 66 revision. That bounded selection is not the complete public corpus. This work proceeds as a focused theory-and-methods candidate; a final ownership decision still requires the wider scientific comparison and evidence that its joint measurement program is independently useful.

2.2 Neural tuning is an inherited scientific topic

Neural tuning is not a term invented by SAN. LaBerge and Kasevich's specific Neuroelectric Tuning hypothesis proposes frequency-tuned apical-dendritic activity in loop circuits [9]. It includes temporal organization and precision, not simply an on/off readiness gate. SAN's integration of that proposal does not transfer authorship of it, and the broader study of neuronal tuning cannot be attributed exclusively to any one recent account.

Kriegeskorte and Wei connect neuronal tuning to population representational geometry and discrimination [10]. That bridge helps ask which differences remain distinguishable at a declared endpoint. It also cautions against inferring a unique cellular implementation from a representational geometry. A description of a population code does not, by itself, specify the physical operation producing it.

2.3 Task-dependent composition already has strong precedents

Cole and colleagues found task-dependent connectivity patterns consistent with flexible, compositional task control [11]. These measurements concern statistical relationships among signals; they do not establish every causal connection required by SAN. Mante and colleagues studied context-dependent selection and integration in prefrontal population activity and a recurrent-network model [12]. Their work is a direct alternative explanation, not merely peripheral background.

Heinen and colleagues experimentally examined how task relevance affects conceptual and perceptual information in human memory [13]. Their behavioral findings motivate testing the accessibility of different representational dimensions. They are not direct measurements of a PWD assembly. Piray and Daw’s computational account of compositional cognitive maps further demonstrates that combining reusable representations for flexible planning is already an explicit research program [14].

Tafazoli and colleagues studied compositionally related tasks in monkeys and found reusable neural subspaces whose engagement varied with the inferred task [16]. Their work makes task inference and subspace reuse concrete alternatives to a hand-specified selector. It does not measure the complete multimodal episode proposed here.

De Oliveira and colleagues studied deep-layer mouse V1 using NREM activity to estimate intrinsic on-manifold, off-manifold and unstructured directions, then compared these to awake movement- and stimulus-related activity [28]. Movement and visual representations interacted additively on-manifold, while visual coding also occupied stimulus-selective off-manifold directions with reduced movement interference. This is a specific biological example of **integration and segregation coexisting in one population**, not evidence that every useful relation should be merged or that task-dependent whole-episode composition has been measured. Its off-manifold terminology is defined relative to their independent intrinsic-dynamics reference; it is not another name for SAN’s PWD. The September source crosswalk separates the conversation-derived lead from this published study.

Mill and Cole studied visual and auditory rule-based task learning in humans, finding a practice-related shift from reusable rule representations toward task-specific conjunctions [17]. Their technical use of compositional versus conjunctive codes must not be confused with my broader use of composition for an episode’s participating contents and relations. Their findings make learning history an essential comparison, not an optional background variable.

Salehi and colleagues combine learned feature processing with top-down and lateral recurrent modulation, task cues, normalization and working-memory components in specified variants [18]. Training for classification and segmentation produces attention and binding behaviors; it is not simply a fixed gain baseline. Their limitations include absent overt eye movements and foveation. A fair comparison must retain those recurrent and relational capabilities. The small GRU comparison below neither reproduces their architecture nor tests superiority over it. Published-model replication remains open.

Consequently, novelty cannot consist of the sentence that a task changes a representation. It would require a precise joint-control model, an independently useful measurement method, or predictive and causal distinctions that these accounts do not already provide. Resource-rational analysis supplies a useful

normative comparator [15], but a normative optimum must not be mistaken for a description of every actual state of a mind.

There is also established work on what a constrained observer can extract. Xu and colleagues define predictive information relative to an allowed prediction family; computation can improve that family’s access without increasing Shannon information [21]. Hewitt and Liang use control tasks to expose how a probe’s own learning affects interpretation [22]. Those contributions predate the present measurement program. My probe accuracy is not an estimate of their information measure. The development-only random-label control added in Section 5.21 is different from their word-type control experiment and is not reported as their selectivity measure.

2.4 From related findings to discriminating controls

The model in Mante and colleagues’ study receives both evidence streams and context [12]. Their population analysis uses targeted dimensionality reduction after denoising, with most biological units recorded in separate sessions. Context-dependent recurrent relaxation selects evidence without requiring removal of the irrelevant stream. The color result in one monkey was equivocal, and the proposed mechanism was not claimed to be exclusive. This is a substantial account of flexible computation, not a fixed-gain caricature. A claim about moment-to-moment effective coupling needs additional evidence beyond a pooled population trajectory.

Tafazoli and colleagues test category decoding across tasks, with controls for stimulus and response variables [16]. Their published task-belief analysis trains on fixation-period activity near block ends and tests earlier discovery periods. Animals inferred the operative task rather than receiving its identity. This matters for the comparison: supplied task cues in my recurrent experiment do not demonstrate learned task inference, and within-role probe accuracy does not establish reuse across roles. These differences motivate the next protocol rather than evidence of superiority over their work.

For the present proposal, the meaningful addition would be to join these established control questions with the distinct contents, bindings, granularity and temporal extent of an episode. A role-specific success is compatible with a reusable code, a reformatted code, selectively accessible information, or a genuinely changed set of represented distinctions. A cross-role probe must therefore be accompanied by within-role and capacity controls, and by consequences of declared interventions. The same reasoning applies to internally generated, bodily and remembered relations; it does not restrict the theory to the visual stimuli used in these comparator studies.

2.5 Human episodes are organized by attended relations and changing state

De Soares and colleagues used audio stories containing overlapping location and social scripts [23]. Location-focused priming shifted subjective event boundaries and mPFC boundaries estimated from multivoxel activity; social priming did not significantly shift annotated boundaries. The positive result is therefore specific, not evidence that every role uniformly reorganizes an episode. It demonstrates an existing human precedent for goal-conditioned temporal organization, not a PWD mechanism. This makes the whole episode, rather than visual acuity alone, an experimentally grounded object of comparison.

McClay and colleagues related music-evoked emotional changes to how neutral image sequences were linked or separated in memory [24]. Emotion ratings were collected during later replay, an important limitation when interpreting state during encoding. This provides a state-sensitive episodic precedent, not evidence that one scalar setting describes every emotion or substance. Neither study tests the complete SAN account.

For the proposed arena comparison, these precedents change the question. A player and a security worker may track different transitions among the same continuing events, not merely remember different static features. An adequate test should ask which event relationships persist across a transition, which become newly consequential, and whether a previously useful organization can be reinstated without replaying the entire environment. Event-boundary judgments alone still do not identify all four configuration coordinates. The added SAN hypothesis must connect those judgments to declared receiver conditions, learned relations and later consequences rather than relabel an established segmentation effect.

Structured Event Memory is a close whole-episode computational comparator [29]. Franklin and colleagues combine structured scene vectors, recurrent event dynamics, event segmentation and reconstruction from noisy memory. Their role/filler representations and event-model reuse already address relational organization and generalization. In my notation, scene constituents overlap with C , role/filler assignments with B , and event membership and within-event dynamics with parts of H . Their noisy distributed representations also constrain accessible distinctions, although this is not the same definition as my finite partition Π . These are correspondences, not claims that SEM implements the four-coordinate controller. My residual question is whether a specified state/task-conditioned *joint reconfiguration* predicts independently measured changes and interventions beyond such an event-memory model, not whether structured episodic reconstruction first appears here.

Hassabis and colleagues’ scene-construction experiment separates described constituents from coherent organization [30]. Five patients with bilateral hippocampal damage and ten controls constructed imagined situations; the group impairment included reduced content and spatial coherence, while one patient was relatively spared. The task scored entities, spatial relations, sensory details and thought/emotion/action, making it relevant beyond visual acuity. In one patient’s assisted trials, providing relevant pictures, sounds and smells did not improve construction. This is a bounded clinical comparison for C -versus- B questions, not a selective manipulation of my coordinates. The small sample, uncertain lesion selectivity, non-blind primary scorer and spatial emphasis preclude identifying the entire (C, B, Π, H) mechanism from this result. The prospective test in Section 6.8 preserves this distinction between constituent access and assembled relations.

2.6 Active acquisition is a substantive existing alternative

EDDI combines partial-observation inference with an expected-information-gain acquisition function [25]. Rahbar and colleagues study online decision-tree learning with active feature acquisition, using a planning oracle and posterior sampling [26]. Valancius and colleagues explicitly address nongreedy acquisition of jointly informative features [27]. Thus learned selection, cost-sensitive querying and the value of complementary observations already have substantial algorithmic precedents. The single-view ablation below cannot stand in for all of that literature.

The new local application is an offline supervised query-tree comparator, not a reproduction of these methods. Its bounded question is what happens when learned coarse/fine requests interact with encountered reference memory, temporary assembly and a later task. It inherits neither the published methods’ guarantees nor their reported performance. This source pass verified the official records and abstracts; full method, software and dataset replication remains open. A distinctive neural claim would require further evidence beyond an application that conventional active acquisition can explain.

3 Composition, granularity, binding, and extent

3.1 A configuration, not a single resolution dial

Let a rendering configuration at time t be described by:

$$K_t = (C_t, B_t, \Pi_t, H_t). \quad (1)$$

Here C_t specifies participating contents or representational variables. B_t specifies their role assignments and relations: which movement belongs to which body, which feature belongs to which object, or which consequence belongs to which action. Π_t specifies distinctions preserved by the representation, operationalized below as a partition in a finite deterministic model. H_t describes represented temporal extent and ordering. These coordinates are analytical distinctions, not a claim that a brain contains four separate control devices.

Composition includes relational organization; B_t is separated to make binding mistakes testable. Extent is not interchangeable with fine granularity. More participating variables do not guarantee more useful information. Redundant or improperly assigned contributions can increase activity without improving a declared endpoint. Conversely, a new relationship can change an action without a larger total signal.

The proposed configuration depends on learned organization M_t , current bodily and other state variables b_t , the operative task g_t , and recent evidence h_t :

$$K_t = F(M_t, b_t, g_t, h_t). \quad (2)$$

This equation names dependencies to investigate; it does not explain them by naming a controller. The candidate implementation is distributed across learned reception, modulation, inhibition, recurrent processing, and sampling behavior. A biological account must identify which of these operations changes which measurable coordinate. It must not posit an unexplained internal observer that selects an already completed experience.

3.2 Granularity is endpoint-relative

For a finite state set S , let a deterministic representation be $r_K : S \rightarrow Z_K$. Its fibers induce the equivalence relation:

$$s \sim_K s' \iff r_K(s) = r_K(s'). \quad (3)$$

This captures distinctions the representation does or does not preserve in the model. It is not a complete definition of neural resolution. Biological measurements are noisy, partial, and time-dependent. Sections 4.8-4.9 now specify a finite stochastic comparison; empirical channel identification, reliability estimation and biological observation windows remain open. A minimum detectable difference along one coordinate remains useful, but it cannot alone describe assembling an episode from many relationships.

The endpoint matters. Distinguishing two orientations, reconstructing who did what, remembering an event order, and selecting a safe action are different demands. A representation can be adequate for one and inadequate for another. A change of task can therefore require a different partition rather than a uniformly finer version of the same partition.

3.3 State changes are not necessarily improvements

Fatigue, sleep, caffeine, alcohol, cannabis, psychedelics, focused practice, meditation, prayer, walking, and gameplay belong in a future program as heterogeneous conditions, not as interchangeable interventions with a common neural signature. This paper reports no substance-specific experiment. A subjective increase in detail might reflect persistence, association, salience, altered sampling, or changed judgment rather than finer sensory discrimination.

The framework is descriptive before it is normative. A configuration can elaborate an irrelevant detail, bind the wrong consequence to an action, or fail to recruit important context. Evaluation must separately measure discrimination, accuracy, scope, temporal organization, confidence, and reported experience. No direction of change in total activity or functional connectivity is assumed.

An odor can make previously background information important to the represented situation. For illustration, a suspected hazardous odor can recruit context, concern for other people, and an evacuation decision. This is not a recommendation to investigate a possible gas leak by remaining in place and sniffing for its source.

3.4 The proposed SAN implementation bridge

A phase-wave differential in SAN is a receiver-relative event difference, not a self-contained memory file [2]. Learned receiving organization, ongoing reference activity, and effective routes condition its consequence. Multiple contributions may become jointly effective without becoming identical or perfectly synchronized. The proposed episode depends on maintaining relevant distinctions while permitting recurrent interaction.

In this account, tuning can change participation and the distinctions a receiver can use. NAPOT names the proposed array-to-array projection and reconstruction operation; a SAN Tomogram describes a reconstructed relation [3]. These concepts locate possible implementation questions, but their names do not establish an end-to-end biological mechanism. A general connectivity change cannot substitute for measuring receiver state, timing, differentiated content, and a downstream consequence.

The continuing action-feedback cycle is essential [5]. Eye and head movements, touch, sniffing in safe experimental settings, posture, and locomotion can change available evidence. Returned evidence can revise both the current rendering and future learned organization. In imagination or remembered episodes, external sampling may be reduced while internally reconstructed relations change. The first finite diagnostic does not implement these temporal operations; the later learned symbolic application tests a restricted two-event case without physiological dynamics.

Consider remembering the arena while no longer being there. Learned associations may make the entrances, particular people, earlier sounds and expected responsibilities available without currently sampling the building. A new question can reorganize that reconstruction around an entrance, a speaker, or the order of events. In a simulation, a changed decoder of a fixed complete memory can imitate some of this behavior. The proposed assembly account therefore needs measurements of what is currently recruited, how relations are assigned, and which distinctions remain available before another retrieval operation. This is a whole-episode question, not merely an adjustable visual sensor.

The existing PWD calculus does not supply a demonstrated generic computational advantage for this implementation bridge. Its stronger synthetic approximation and reconstruction comparisons did not establish the proposed advantage over ordinary dynamic alternatives [2]. Those reported results concern particular implementations and tasks; I do not generalize them to every possible receiver-relative model. Conversely, invoking the broader SAN architecture here cannot turn those outcomes into evidence for this

paper. The new learned application must make its own information, memory and intervention boundaries explicit.

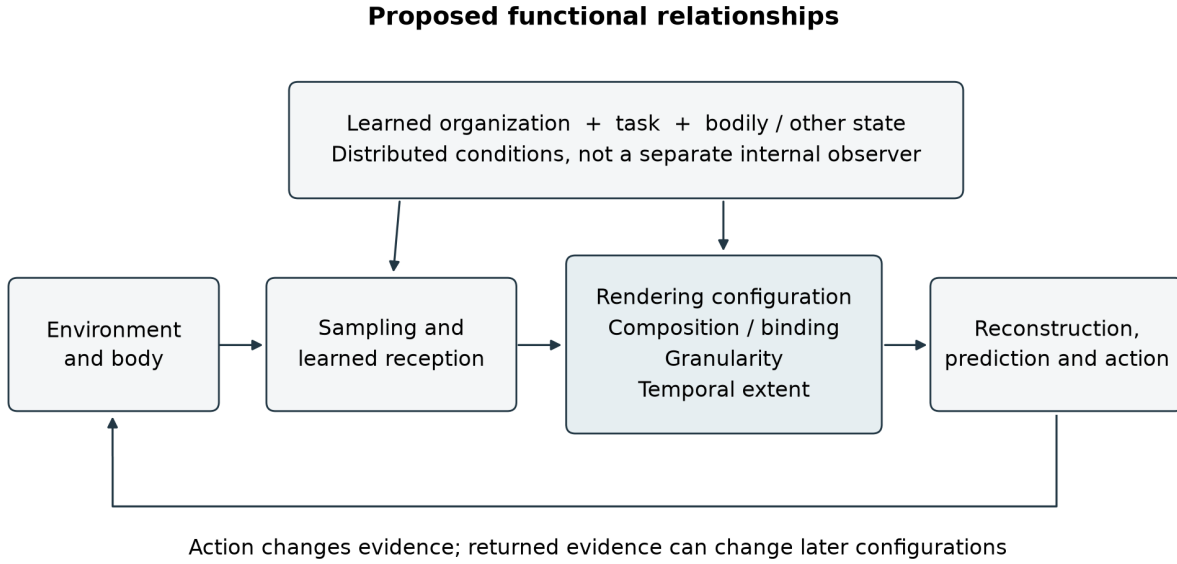


Figure 1: Proposed control relationships. The organism interacts with an environment while learned organization, task and state condition a rendering configuration. The lower path shows action-dependent returned evidence. Arrows are functional hypotheses, not measured neural pathways.

3.5 Reconfiguration is a transition, not a completed episode choosing itself

The dependency statement can be extended to the transition at issue. This step instantiates the earlier dynamic-canvas and tuning-update questions [1,2]; writing a next-state equation is not itself the contribution. A minimal specification is:

$$K_{t+1} = \mathcal{U}(K_t, M_t, b_{t+1}, g_{t+1}, e_{t+1}). \quad (4)$$

Here e_{t+1} is evidence available after the previous sampling and action, not an omniscient description of the environment. The previous action can change which evidence arrives. Learned organization can also change on its own relevant timescale; it should not be equated with the instantaneous configuration. The function $\mathcal{U}$ is an unknown to identify, not an explanation supplied merely by writing the equation.

For example, a person managing an arena may receive a cue that a particular entrance needs attention. The proposed transition can preserve the larger event’s remembered responsibilities while refining a local relation, changing which sound or movement matters, and revising the interval over which consequences are anticipated. It need not increase every dimension or discard all preceding context. The later return to event management is part of the same testable sequence.

A measured change in behavior alone does not locate that transition. It could arise from a new readout of unchanged available information, new external sampling, a genuinely altered internal assembly, or a combination. Section 4.5 formalizes one ambiguity. The transition model earns explanatory value only when these alternatives are constrained by measurements or interventions.

4 Elementary mathematical guarantees and counterexamples

The following propositions are elementary consequences of finite representations and unrestricted readouts. They are included to expose assumptions and make the operational model checkable. They are not claims of new mathematical discoveries, nor proofs that a brain implements SAN.

4.1 A task can be read out exactly when its distinctions survive

Let S be a nonempty finite set, let $q : S \rightarrow A$ be a declared target, and let $r : S \rightarrow Z$. Consider readouts defined on the attainable image $r(S)$.

Proposition 1 (task sufficiency). A readout $d : r(S) \rightarrow A$ satisfying $q = d \circ r$ exists if and only if q is constant on each fiber of r .

Proof. If $r(s) = r(s')$, any such readout gives $q(s) = d(r(s)) = d(r(s')) = q(s')$. Conversely, if the target is constant on each fiber, define $d(z)$ to be the target of any state mapped to z . Constancy makes this choice well-defined and establishes the factorization. This concerns exact target access, not phenomenal experience.

For uniformly weighted states and zero-one loss, the optimal readout error is:

$$L^*(r, q) = 1 - \frac{1}{|S|} \sum_{z \in r(S)} \max_{a \in A} |\{s \in S : r(s) = z, q(s) = a\}|. \quad (5)$$

For each representation value, the best deterministic readout chooses a most frequent target in that fiber. Summing these correct counts proves the expression. Randomizing a readout cannot exceed the maximum frequency within a fiber. This exact finite calculation is the evaluator used in Sections 5.1-5.2; it is not a trained or held-out decoder. The later learned application declares a different task head and a separate completion diagnostic.

4.2 Refinement helps an unrestricted readout, but can have a cost

Proposition 2 (refinement). Suppose $r_c = h \circ r_f$, so the coarse representation is recoverable from the fine one. Under the same state distribution and target, with bounded loss and a finite nonempty readout action set, and allowing all readout maps, the minimum readout loss obeys:

$$L^*(r_f, q) \leq L^*(r_c, q). \quad (6)$$

Proof. Every coarse readout d_c has a fine readout $d_f = d_c \circ h$ with identical predictions and loss. The fine representation’s minimum cannot be worse than this available choice. A restricted or unsuccessfully trained decoder need not achieve this guarantee.

The finite action set ensures that a minimum is attained. With an infinite action set, the corresponding inequality holds for infima; the existence of a minimizing readout needs a separate assumption.

The proposition does not say that maximal detail is always useful. If c_f and c_c denote declared acquisition or processing costs, a fine representation is preferred under loss-plus-cost only when:

$$L^*(r_c, q) - L^*(r_f, q) > \lambda(c_f - c_c). \quad (7)$$

Here $\lambda \geq 0$ weights the declared cost. This is an explicit normative comparison, not evidence that neurons literally evaluate the inequality. Costs must include the selection and control machinery when making system-level efficiency claims.

4.3 Task-specific partitions need not form one resolution ladder

Proposition 3 (incomparability). On $S = \{0, 1\}^2$, take $r_A(x, y) = x$ and $r_B(x, y) = y$. Neither representation determines the other.

Proof. States $(0, 0)$ and $(0, 1)$ have identical r_A but different r_B , so no function of r_A recovers r_B . States $(0, 0)$ and $(1, 0)$ give the reverse obstruction. Each representation is sufficient for its corresponding target, but neither is simply higher resolution than the other. A representation adequate for both must distinguish both coordinates, or achieve the same joint distinction by another encoding.

This small counterexample is why composition cannot be reduced to the number of distinguishable states. Two representations with the same cardinality can preserve different action-relevant distinctions. It does not imply that every biological pair of task representations is incomparable.

4.4 Gain and binding require explicit observation assumptions

An invertible transformation of a noiseless representation preserves its fibers, and therefore preserves optimal readout performance. Multiplying finite codes by two cannot create distinctions that their original encoding discarded. But gain can improve discrimination when it precedes noise. For equally likely labels $Y \in \{-1, 1\}$ and $X = gY + \epsilon$, with independent standard Gaussian noise and $g > 0$, the optimal sign readout has accuracy:

$$P(\text{sign}(X) = Y) = \Phi(g). \quad (8)$$

Thus the analytic accuracy increases from approximately 0.8413 at gain one to 0.9772 at gain two. The claim that gain can never improve resolution would be false. Any experimental separation must declare where gain, noise, sampling, and readout operate.

Similarly, swapping two labeled slots can cause a fixed readout to fail even though an invertible permutation preserves all information. That is a binding or interface mismatch relative to the declared readout, not information destruction. A compensating decoder can undo it. Section 5 uses this limited perturbation transparently.

4.5 The same task behavior need not identify the same assembly

Proposition 4 (nonidentifiability from the declared primary tasks). There exist task-conditioned and task-independent encoding rules with different representation fibers that produce identical correct behavior on every sequence of the declared primary tasks.

Proof by construction. Let $S = \{0, 1\}^3$, write $s = (x, y, z)$, and let the two tasks be $q_0(s) = x$ and $q_1(s) = y$. A selective encoder retains $r_g(s) = q_g(s)$, with identity readout. A full encoder retains $r_F(s) = s$ regardless of the task; its readout d_g selects the requested coordinate. Therefore, for every state and task,

$$\text{id}(r_g(s)) = d_g(r_F(s)) = q_g(s). \quad (9)$$

Applying this pointwise identity at each step gives identical outputs for arbitrary state/task sequences. Yet for fixed g , the selective encoding has two attainable codes and four states per fiber; the full encoding has eight singleton fibers. Thus the observed primary outputs do not identify which distinctions the encoding retains. This does not say that every experimental observation is uninformative or that the two complete systems are otherwise equivalent.

An auxiliary query about z , applied to the **unchanged** representation under a uniform independent state distribution, has optimal accuracy one half for selection and one for the full encoding. The construction therefore also supplies a possible discriminating observation under stated access conditions. In a human experiment, however, issuing a probe can cause renewed retrieval, sampling, or reconfiguration. That is not automatically the readout-only intervention assumed by the proof.

The encoding rule in the full model is task-independent, but its values still change with the observed world. Neither model in this construction maintains temporal memory. Furthermore, an invertible recoding with a compensating decoder preserves all fibers and primary outputs. Coordinate labels cannot uniquely identify a biological implementation. These are familiar identifiability issues applied to this paper’s measurement problem, not new general principles of causal inference.

4.6 Current-task sufficiency does not guarantee every future answer

Proposition 5 (unrestricted query preservation). Fix the state space S and the unchanged representation $r : S \rightarrow Z$. Every binary query $q : S \rightarrow \{0, 1\}$ can be read out exactly from $r(S)$ if and only if r is injective.

Proof. If two distinct states s and t have the same representation, choose the binary query that is zero at s and one everywhere else. It differs at t , violating Proposition 1. Therefore exact access to every such query requires that no distinct states be merged. Conversely, an injective representation has singleton fibers, so every binary query is constant on its fibers and admits an exact readout. This is a familiar consequence of information-preserving encoding, not a new general mathematical discovery.

The distinction matters when the task changes. A selected assembly can preserve everything required for the present decision while omitting distinctions needed for another one. Answering a newly relevant question can therefore require a new assembly from stored evidence or a new observation. Those operations change the available representation; they do not violate the frozen-representation result. Section 5’s separation of event memory, current assembly and renewed sampling makes this boundary executable.

The quantifier is deliberately strong: *every binary query on the declared model states*. It is not a claim that useful organisms must preserve every microscopic fact, that all future tasks are relevant, or that ordinary memory is lossless. A restricted task family can permit substantial compression. Lossy memory, finite effort and uncertain observations remain realistic possibilities. The result identifies what a guarantee would require, not what brains are assumed to do.

4.7 What the formal supplement verifies

The accompanying Lean 4.30.0 source contains twenty compiler-checked statements: general factorization and collision results; simulation of coarse readouts from fine representations; risk ordering conditional on an attained fine-readout minimum; invariance under a left-invertible recoding; Proposition 5; binary incomparability and binding-rescue examples; the selective/full primary-behavior construction and its distinguishing auxiliary query; and the finite temporal-order count and upper bound used below. The successful compiler check returned exit code zero. The reported dependencies contain no admitted-proof axiom or custom domain axiom; some proofs use Lean’s standard classical choice, propositional

extensionality and quotient-soundness principles. The mathematical premises stated in each theorem still apply.

The formal risk theorem makes the existence of an optimal fine decoder an explicit premise. It verifies the simulation argument in Proposition 2, not the separate finite-minimum existence argument or a learning algorithm’s ability to attain it. Likewise, the symbolic constructions for Proposition 4 are verified, but its uniform-probability accuracy discussion is not a formal probabilistic theorem. The general maximum-frequency evaluator, Gaussian-noise example and total-cost comparison retain their analytic proofs rather than being described as Lean results.

The twenty statements are a reproducibility inventory, not twenty independent discoveries. The code does not formalize neurons, PWD dynamics, a stochastic temporal process, consciousness, or the Python application’s execution. Compiler verification is separate from independent scientific or formal-methods review, which remains pending. Failed compiler and environment attempts are preserved alongside the successful source and axiom audit.

4.8 Stochastic observations require an explicit comparison

Exact fibers do not capture noisy reception or incomplete recording. For this extension, fix a finite state set, finite nonempty observation and action sets, a state prior π , and a common loss $\ell(s, a)$ in $[0, 1]$. An observation channel $E(z \mid s)$ is a nonnegative row-stochastic matrix. It describes the distribution of recorded evidence given a model state, not a biological mechanism merely by notation. The task, bodily context and observation time are held fixed for a comparison or declared explicitly in separate conditional analyses. Write $R_\pi(E)$ for minimum expected loss over all readout rules. Finiteness guarantees an attained optimum.

Proposition 6 (finite stochastic postprocessing). If a second channel F can be obtained from E by a row-stochastic transformation T that has no additional access to the state, then:

$$F(y \mid s) = \sum_z E(z \mid s)T(y \mid z) \implies R_\pi(E) \leq R_\pi(F). \quad (10)$$

Proof. Given an optimal rule for F , a receiver of E ’s output can sample T and apply that same rule. By the displayed identity, the resulting joint distribution of state and action, and hence expected loss, is identical. This is an available randomized rule for E . At each observation, its conditional loss is a convex combination of the losses of individual actions, so choosing a minimizing action gives a deterministic rule no worse. The minimum for E therefore cannot exceed that for F . This is the elementary forward postprocessing argument of statistical decision theory, not a newly claimed general theorem or a proof of its converse.

The assumption is stronger than observing one lower accuracy. A declared transformation must reproduce the conditional channel for every state being compared. Two independently trained networks are not known to satisfy this assumption. Different noise, task priors, cues or targets also invalidate an unqualified comparison. For a restricted readout family, the simulation argument only applies when the allowed fine rules can implement the transformation and coarse decision, or an equally effective substitute. Learning may fail even when that composition is representable.

4.9 Approximate channels and endpoint-specific resolution

For two distributions on the same finite alphabet, define total variation as half the sum of their absolute probability differences. If $\tilde{F} = ET$ approximates F , let ϵ be the prior-weighted total variation between their rows. Then:

$$\epsilon = \sum_s \pi(s) \text{TV}(F_s, \tilde{F}_s), \quad |R_\pi(F) - R_\pi(\tilde{F})| \leq \epsilon. \quad (11)$$

Proposition 7 (finite bounded-loss stability). Under the preceding common prior, common action set and loss in $[0, 1]$, the displayed bound holds. Together with Proposition 6 it implies $R_\pi(E) \leq R_\pi(F) + \epsilon$.

Proof. Fix any common readout. Its loss, conditional on each observation and state, lies in $[0, 1]$. The absolute difference of its expectations under two row distributions is at most their total variation: positive and negative probability differences have equal total mass, and a function in $[0, 1]$ cannot amplify either mass. Average over π . Use an optimal rule for either channel as an available rule for the other to obtain the two directions of the bound on minima. Finally, Proposition 6 gives $R_\pi(E) \leq R_\pi(\tilde{F})$, which yields the stated approximate ordering.

This is a bound for known probability tables, not an estimator or confidence interval. Establishing it empirically requires a justified bound on the conditional observation model. The result does not apply to unrestricted log loss, which is unbounded near zero predicted probability. In particular, it cannot be applied directly to the recurrent experiment's reported negative log likelihood. Nor does a small error under one prior ensure a small error under another prior that emphasizes previously rare states.

For an explicitly defined binary discrimination endpoint with equal class priors and conditional observation distributions P_0 and P_1 , unrestricted optimal accuracy is:

$$p^* = \frac{1}{2} \sum_z \max\{P_0(z), P_1(z)\} = \frac{1 + \text{TV}(P_0, P_1)}{2}. \quad (12)$$

The first equality chooses the more likely class at each output. The second follows by substituting $\max(a, b) = (a + b + |a - b|)/2$ and using that each distribution sums to one. To call a difference resolvable, one must declare the alternatives, their priors, recording/noise channel, readout restrictions and an accuracy criterion. If a continuous family is used, an infimum detectable difference need not be attained or vary monotonically; there may be no qualifying difference at all. No universal scalar resolution follows from this binary result.

4.10 A lossless change can defeat every affine threshold

Let (u, v) range uniformly over $\{-1, 1\}^2$, with the target v . Compare the identity encoding with the recoding $f(u, v) = (u, uv)$. The recoding is invertible: multiply its two output coordinates to recover v , while its first coordinate already gives u . Every original distinction therefore remains available to an unrestricted decoder. Both encodings have optimal target accuracy one.

An affine threshold on the identity encoding can read v perfectly. Under f , the positive-target states are the two diagonal corners and the negative-target states the other two corners. Both pairs have the origin as their midpoint. For a rule that labels nonnegative affine scores positive, perfect positive classification requires nonnegative scores at both positive corners, so their average, the intercept, is nonnegative. Perfect negative classification requires strictly negative scores at both negative corners, so that same intercept is negative.

This is impossible. Reversing the tie convention gives the same obstruction with reversed strictness. A line can classify three corners correctly, making the affine optimum exactly three quarters; a product readout restores accuracy one.

This is not the earlier example of keeping an inappropriate *fixed* decoder. Even the optimal member of an entire affine family is worse after a lossless recoding. The example proves possibility, not the cause of the recurrent networks’ observed probe ranking. It fits the established distinction between unconstrained information and access through a restricted predictor [21]. Low affine accuracy can be a meaningful failure for an affine receiver, but it does not establish erased memory or a universally coarser rendering.

Propositions 6-7 and the binary and affine examples are hand-derived results with finite rational checks below. The existing twenty Lean statements remain unchanged and do not formally verify these additions. Their role is to specify the measurement problem and prevent inappropriate inference from the applications, not to add a count of purported new mathematical discoveries.

5 Executable diagnostics and learned applications

5.1 Design and reproducibility

The reference application enumerates six independent four-valued coordinates, giving 4,096 equally weighted states. The coordinates carry illustrative names: visual geometry, body state, auditory sequence, remembered context, conceptual relation, and chemical identity. These are symbolic integers. They are not recordings, learned sensory representations, realistic memories, or a multimodal foundation model.

Six declared roles request ordered pairs of coordinates: player requests geometry and body state; security requests auditory sequence and remembered context; manager requests remembered context and conceptual relation; speaker requests conceptual relation and auditory sequence; odor response requests chemical identity and remembered context; food handling requests chemical identity and body state. Every role has 16 equally frequent target pairs.

Eight encoding conditions are evaluated under each role. The role-conditioned assembly preserves the two requested coordinates at their original four-valued granularity. An ordinary task-attention baseline uses exactly the same selection. Fixed fine and fixed gain conditions preserve the first two coordinates, with the latter applying an invertible rescaling. A fixed coarse condition retains the first four coordinates after merging each pair of adjacent values. A fixed low condition retains only the first two coarse coordinates. A full condition retains all six fine coordinates. A stale-role condition applies the previous role’s selector in a declared cyclic role order.

The evaluator uses the exact optimal-fiber readout from Section 4. There are 48 conditions and 196,608 state-condition evaluations. Those are deterministic evaluations over one constructed universe, not independent empirical observations. The code does not learn a selector, train a decoder, infer a hidden task, or test generalization to unseen states. Reported bit counts describe retained coordinate alphabets only; instructions, role labels, routing, code, and computation are excluded.

5.2 Results, including the attention tie

Averaged equally across the six roles, the hand-specified assembly and the ordinary attention baseline both achieve accuracy 1.000 using four retained source bits. The full fine representation also achieves 1.000 while retaining twelve source bits. The fixed fine pair and its gain-rescaled counterpart each achieve 0.250. Fixed

coarse four-coordinate encoding achieves 0.1667; the fixed low pair achieves 0.1042. The stale previous-role selector achieves 0.1875. Figure 2 reports the exact finite-model results, rounded for display.

The strongest conclusion is limited: a role-aware selection preserves the target coordinates in this constructed problem, while a mismatched fixed selection often does not. This behavior follows directly from the task definitions. Ordinary attention accomplishes it equally well. The diagnostic therefore provides no evidence that the new terminology identifies a superior algorithm, and no proof that four source bits imply a threefold reduction in total system cost.

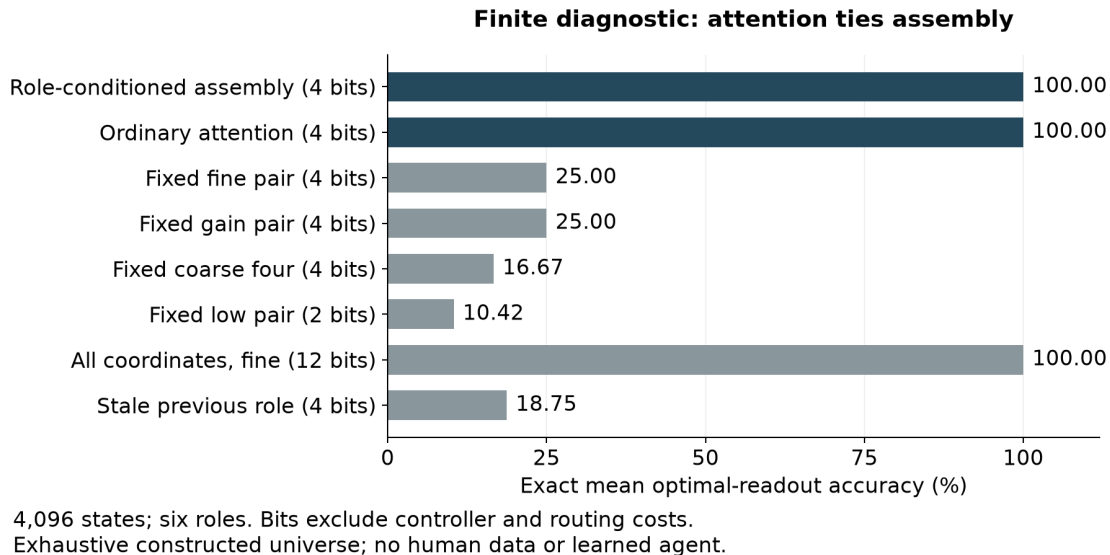


Figure 2: Exact mean optimal-readout accuracy across six declared roles. The assembly and ordinary-attention encoders tie by construction. Parenthetical bit counts measure retained source coordinates only. No sampling error bars are applicable to exhaustive evaluation of this fixed universe.

A separate fixed-readout check swaps the two requested slots. The original identity readout is correct only when both values match, giving accuracy 0.25. Restoring the slot assignment restores accuracy 1.00. Because swapping is invertible, this is not loss of Shannon information. Its purpose is to separate content availability from compatibility with a particular downstream interface.

Eight executable consistency checks pass. They verify the attention tie, invertible-gain invariance in the noiseless fixture, full-representation accuracy, the tested refinement inequality, the binding perturbation and rescue, an incomparability witness, and exact target balance. The Gaussian gain example is an analytic calculation, not a second neural experiment. The formal supplement verifies the specified representation/readout statements described in Section 4.7; it does not certify this application’s code or every numerical claim.

5.3 What the first diagnostic does not implement

There is no acquired memory, temporal episode, changing body, noisy receptor, neural morphology, PWD timing, active sampling, recurrent plasticity, or report of experience. The stale selector is a deliberately mismatched route, not an experimentally grounded model of fatigue or intoxication. The application is a

reproducible specification test and teaching instrument. Calling it proof of a working biological rendering mechanism would exceed its scope.

5.4 A second diagnostic separates performance from retained information

The second program implements Proposition 4 on eight binary-coordinate states and two roles. It exhausts all three-step state paths and role paths: 4,096 paired trajectories and 12,288 paired step comparisons. The selective and full encoders give identical, correct primary responses throughout. These counts describe exhaustive constructed cases, not independent empirical samples.

For each role, the program then freezes the representation and asks three readout-only questions: the current target, the other role’s target, and the unused coordinate. Optimal accuracies are respectively 1.00, 0.50, and 0.50 for selection, and 1.00 for all three questions under the full encoding. Figure 3 displays these values averaged equally across the two roles. A separate invertible-recoding check preserves primary performance.

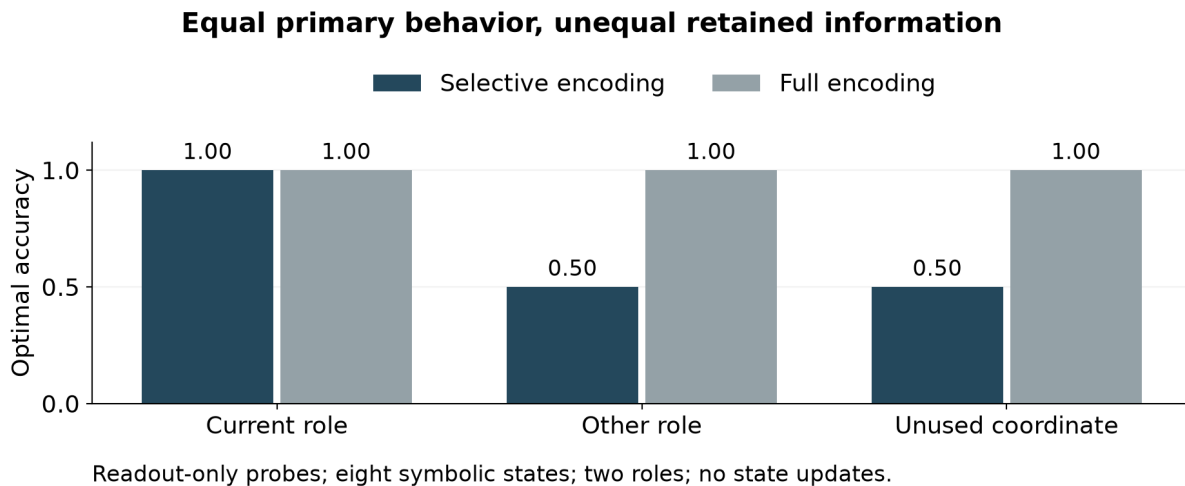


Figure 3: Identical primary-task behavior does not identify all retained distinctions. Readout-only probes of the frozen artificial state distinguish these encoders. Values are exact finite-universe accuracies; there are no sampling-error bars. A probe of a living person is not assumed to freeze neural state or prevent further retrieval.

All 17 checks pass. The adjacent protocol states the analytic expectations before execution. The result is deliberately designed to expose an ambiguity, not to discover a neural effect. It adds no learning, realistic memory, PWD dynamics, or biological cost claim to the first application.

The finding changes the experiment’s burden of evidence. Better role-specific performance is not by itself evidence for a changed internal assembly; the measurement must constrain what remains available under a declared state boundary. A frozen snapshot of a completely specified artificial system makes that intervention possible. Partial access to a human brain, or to only one module of an artificial agent, does not automatically provide the same boundary.

5.5 A learned episode application: what is acquired and what is supplied

The third program implements a small learned controller rather than another hand-selected encoder. Its world contains three entities with two four-valued channels each. An opaque channel codebook converts values into observation symbols. Labeled calibration encounters drawn from training scenes teach twelve symbol meanings: eight fine meanings and four coarse-bin meanings. These acquired associations are distinct from the episodic record of subsequently encountered, entity-bound observations. A temporary assembly selects records from that episode, and a separate fixed task head reads the assembly. This is supervised symbolic learning, not learned sensory physiology or a neural memory model.

There are 250 programmer-defined plans. Each chooses one, two or three coordinates, or the complete six-coordinate set; chooses coarse or fine representation; and requests current samples, reference-event retrieval, or both. Plans requiring more than six operations are excluded. Coarse observations merge adjacent values; fine observations retain four values. The controller learns mean rewarded utility for each plan under each task cue and stage by balanced exploration of 32 training episode pairs. Utility is response accuracy minus declared acquisition-bit, retained-bit and operation penalties. It learns which macro-action plan to use. It does not learn the plan grammar, task semantics, a neural representation, or within-query micro-action sequencing.

Every episode begins with six explicit fine observations of the reference scene, identically supplied and charged to every condition. This controlled familiarization makes earlier evidence available; it is not learned anticipatory acquisition. A first task uses that scene. The world then changes, a different task arrives, and the second response may require new samples, a remembered relation, or both. The six tasks request a fine value from coordinate 0 or 5, a broad coarse relation across coordinates 2-4, a correctly bound pair in coordinates 1 and 3, a delayed pair in coordinates 0 and 5, or the direction of change in coordinate 4. These abstract channels and tasks should not be confused with realistic human occupations or experiences.

For each of three fixed seeds, the 4,096 instantaneous states are partitioned into 2,867 training, 614 development and 615 evaluation states. Every marginal symbol appears in training. The application fits associations and 2,500 stage/task/plan utility estimates per seed, freezes those parameters, then constructs evaluation episodes. There are twenty task-pair orders and four repetitions per seed: 240 paired two-query episodes, evaluated under eleven conditions, yielding 5,280 query-condition rows. Those rows are not independent empirical observations. Evaluation scenes and complete sequences do not overlap the fitting scenes and sequences. No reward updates the learned parameters during evaluation. The generator, plans and endpoints were authored for this development program; withheld combinations are not independent-generator validation.

5.6 Learned plans, intervention results and the surviving alternatives

All three seeds learn the same selected plans. The first query retrieves the relevant reference records. After the switch, fine single-coordinate tasks request one current fine sample; broad coarse context requests three current coarse samples; binding requests two current fine samples; the delayed pair retrieves two old records; and event order combines one old record with one new fine sample. The controller receives an explicit task cue, not a hidden answer or an inferred human intention. Sampling returns only the requested coordinate. Retrieval returns only previously observed records.

The learned controller answers all 480 queries correctly. An equivalent attention implementation with the same fitted controller, memory and readout produces identical assemblies, actions and answers. It is an interpretation control, not an independently trained state-of-the-art comparator. An information-rich full-record baseline also answers all queries correctly, but receives all six current fine samples and the archive. It

is not a matched-acquisition-budget competitor. A fixed all-coarse sampler is also recorded, along with the interventions below. No predictive superiority over a strong recurrent model is established.

Figure 4 shows second-query results. Restricting the assembly to the current event leaves all four current-scene tasks correct, while the original readout drops to 1/48 on the delayed pair and 20/48 on event order. Deleting prior memory produces the same primary responses, but a different underlying memory state. The event ledger distinguishes those causes; behavior alone does not. Denying new samples instead preserves the delayed-pair response while disrupting current-scene tasks. Permuting entity bindings leaves the values present but disrupts their fixed interface; applying the inverse map restores the exact assembly and its responses. Reversing time labels preserves records but changes their interpreted order. These are controlled software interventions, not selective manipulations of human brain circuits.

After the task switch: observed fixed-head accuracy

Learned plans	1.00	1.00	1.00	1.00	1.00	1.00
Equivalent attention	1.00	1.00	1.00	1.00	1.00	1.00
Full fine record	1.00	1.00	1.00	1.00	1.00	1.00
Assembly coarsened	0.25	0.22	1.00	0.03	0.02	0.42
Current-only assembly	1.00	1.00	1.00	1.00	0.02	0.42
Prior memory reset	1.00	1.00	1.00	1.00	0.02	0.42
No new samples	0.25	0.22	0.06	0.03	1.00	0.42
Bindings permuted	0.25	0.22	0.06	0.03	0.02	0.42
Inverse binding rescue	1.00	1.00	1.00	1.00	1.00	1.00
Time labels reversed	0.25	0.22	0.06	0.03	0.02	0.25
	Fine 0	Fine 5	Broad coarse	Bound pair	Past pair	Event order

3 synthetic seeds; 36 or 48 paired second queries per task and condition

Figure 4: Observed accuracy after the task switch in the learned symbolic application. Values aggregate three seeds; there are 36 second queries for each current-scene task and 48 for each delayed/order task per condition. The attention control is explicitly equivalent, and the full fine record receives more source information. Failure of the fixed head does not quantify all retained information; Figure 5 applies a different readout.

At the second query, the learned plans acquire 2, 2, 3, 4, 0 and 2 source bits for the six tasks respectively. Their temporary assemblies retain 2, 2, 3, 4, 4 and 4 coordinate-value bits. These different quantities separate newly sampled evidence from retained historical information. Every episode also incurs twelve familiarization source bits, which must not disappear from total acquisition accounting. The full-record baseline acquires twelve new source bits and retains twenty-four coordinate-value bits at the second query. These alphabet counts omit entity/time labels, metadata and implementation overhead; actual serialized

record sizes, operation counts, table lookups and 80,000 training plan/episode evaluations per seed are logged separately. None is labeled metabolic energy or total neural compute.

5.7 A failed fine readout does not imply empty coarse information

The original task head requires explicit fine values for fine queries; when an entry is missing or only coarse, it emits a fixed zero response. That declared rule simplifies the control problem but does not optimally exploit partial evidence. It would be an error to interpret every failure after coarsening as complete information loss. I therefore added a clearly post-hoc completion diagnostic: keep the assembly frozen, enumerate the fine values consistent with its recorded bins or missing entries under an independent uniform-coordinate prior, and select the most probable task answer with fixed tie-breaking. This alternate readout has no access to the archive, new samples or true target. It is not a newly preregistered endpoint or a posterior fitted to the restricted evaluation pool.

After assembly-only coarsening, both readouts remain correct on all 36 broad-coarse queries. For the two fine single-coordinate tasks, the original head scores 9/36 and 8/36; the completion readout scores 16/36 and 19/36. For the bound and delayed pairs it improves from 1/36 and 1/48 to 7/36 and 9/48. For event order it improves from 20/48 to 38/48. Figure 5 displays this difference. The coarse representation has lost some fine distinctions, but a head that ignores its bins exaggerates the practical loss. Likewise, an invertible binding mismatch is not erasure: a decoder that knows the inverse can recover the original relations.

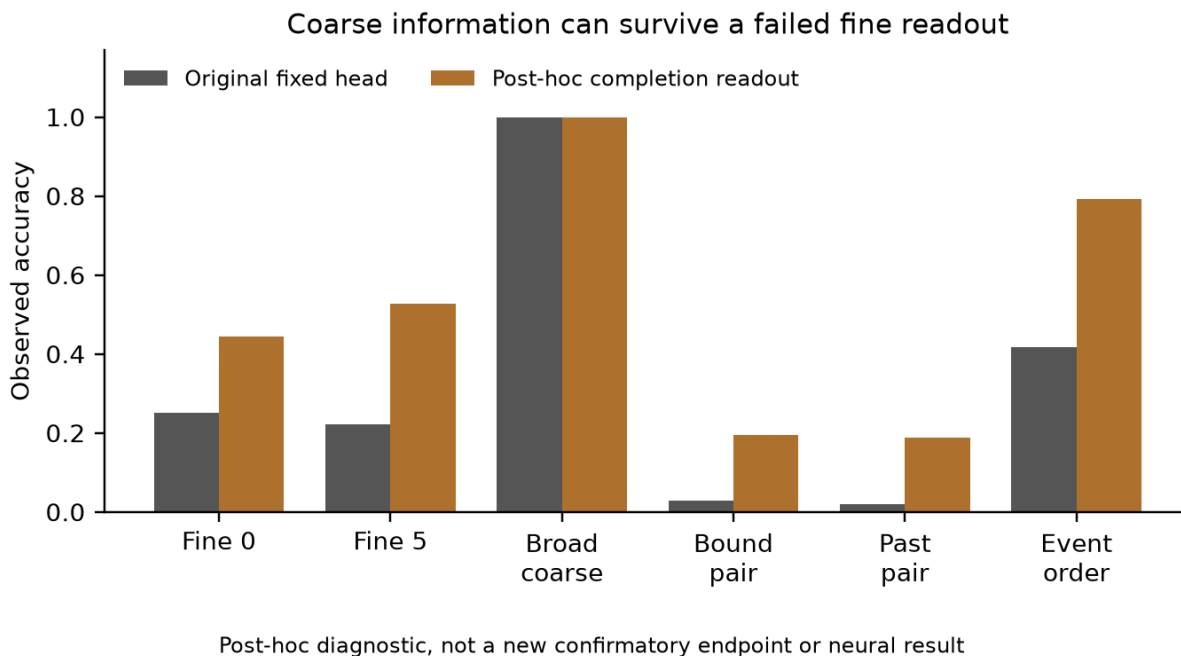


Figure 5: A post-hoc alternate readout of the same coarsened assemblies often recovers information that the original fine-response head did not use. Both readouts score the identical recorded evaluation cases. The uniform-completion assumption and post-hoc status are explicit; these are descriptive development results, not a neural effect or a calibrated biological posterior.

A small exact consequence of the partition model makes the distinction transparent. Let old and new values be independent and uniform on four ordered values, and let each observation retain only its adjacent two-value bin. Of the sixteen equally likely ordered pairs, eight lie in different bins and reveal increase versus decrease exactly. In each same-bin group of four, equality occurs twice and each direction once, so equality is the most probable answer. The optimal expected three-way order accuracy is therefore

$$\text{Acc}_{\text{coarse order}}^* = \frac{8 + 2 + 2}{16} = \frac{3}{4}. \quad (13)$$

This is a worked finite consequence of task sufficiency and fiber-wise optimal readout, not a new general theorem. All sixteen cases were checked independently. The 38/48 observed diagnostic score above is a finite result on a different restricted scene sample; three quarters is the expectation under the stated uniform model, not a ceiling on every finite sample’s accuracy. A claim about lost distinctions must name its observation model, allowed decoder and distribution.

5.8 Reproduction and implementation boundaries

Twenty-one unit checks pass. Twenty-two subsequent checks independently recompute all logged target labels and scores, verify splits and freeze ordering, inspect intervention identities and costs, and reproduce the complete training and all 5,280 evaluation rows. Replayed query records match the stored serialization byte for byte. The first verifier pass exposed a tuple-versus-list comparison mismatch after JSON serialization; exact serialized bytes already matched. The corrected comparison uses the declared JSON data format, and the unsuccessful verifier receipt is preserved. No application parameters or results changed to resolve that check.

The initial smoke test used development states only. The complete three-seed run used one below-normal-priority CPU process, took about 1.8 seconds on the local machine and wrote approximately 5.2 MB of compact results and parameters. Those measurements describe this execution, not a general speed benchmark. The runnable package includes the frozen configuration, protocol, source, trained tables, per-query actions and assemblies, hashes, examples and readout diagnostic. It contains no model download, neural recordings, pharmacology or report of experience.

The application is now learned in two specific respects: symbol associations and task-conditioned plan values are acquired from training. Its event memory stores actual observations and affects later answers. But the task heads and available macro-actions are supplied, the reference scene is explicitly familiarized, and the same simple generator supplies both training and evaluation. It cannot establish how a brain learns rich episodes, how PWD contributions are physically assembled, or whether the proposed coordinates add predictive value beyond modern recurrent attention. Those are next-stage burdens rather than properties inferred from a perfect symbolic score.

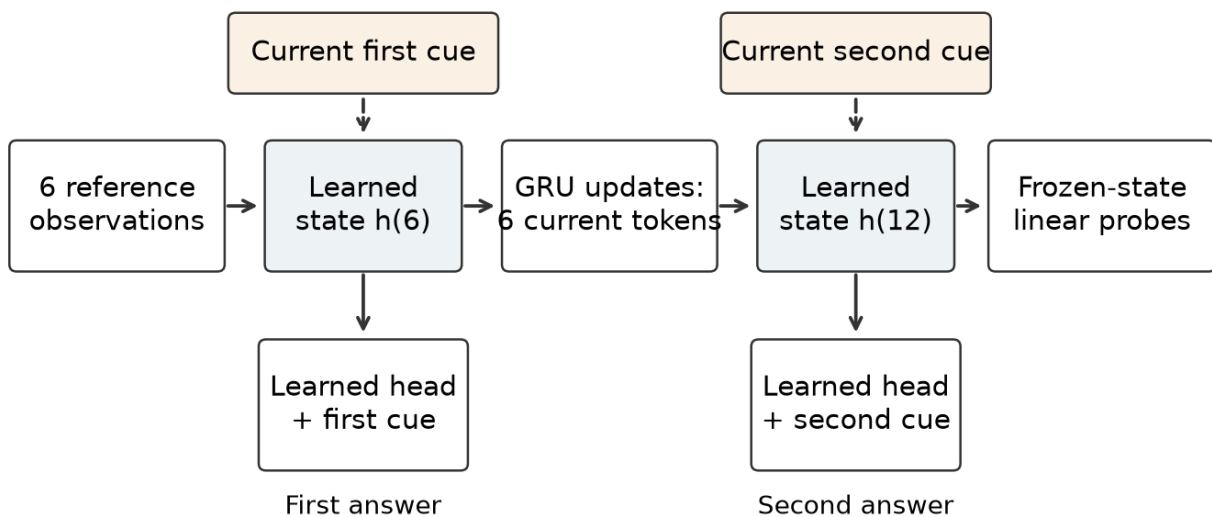
5.9 A recurrent comparison learns both its representation and answer rule

The fourth program replaces supplied answer rules and explicit event records with a trained recurrent representation and nonlinear readout. It uses the established gated recurrent unit family [20], implemented in NumPy for inspectability. This is an ordinary neural-network control, not a newly invented SAN architecture. The two conditions share a 48-unit GRU and a 64-unit nonlinear answer head with a common sixteen-class output. Both receive identical fine observations and the current task cue at the answer head. Only one receives that cue during recurrent encoding. The head branches from the state and cannot write back into memory (Figure 6).

Each episode again has two scenes of three entities with two four-valued channels. Twelve randomly ordered tokens identify entity, channel, scene and value. There are six tokens per scene and an answer after each scene. The task changes between scenes; the future task is not disclosed during the first scene. Targets retain the six definitions in Section 5.5, but the network receives neither target coordinates nor the formula for an answer. Supervised examples teach that mapping. There is no legal-output mask that supplies a task-specific answer range. These remain symbolic channels, not audiovisual recordings or multimodal experience.

The two conditions have 14,064 allocated parameters, matched initial arrays per seed, identical episodes, minibatch order, optimization settings and epoch budgets. Their information access differs intentionally: the readout-only condition has zero encoder cue columns. Those columns contain 864 allocated but unused weights across three gate matrices. Equal allocated size therefore does not mean identical effective parameter use or equally available control information. The contrast tests cue access during encoding under this training procedure; it is not an equal-information contest proving algorithmic superiority.

Learned recurrent comparison: encoding versus readout cue access



Dashed cue access differs. Both heads share learned weights; neither writes back. Same 48-unit GRU, fine observations and training budget. Future cue is not disclosed early.

Figure 6: Recurrent application architecture. The dashed cue access distinguishes the conditions; both answer heads receive their current task cue and share learned parameters. The reference state carries into GRU updates on current tokens. Observation schedules are fixed, so the arrows do not denote learned acquisition. Frozen-state probes are separate trained readouts, not recurrent feedback or measurements of a human brain.

5.10 Training, held-out evaluation and the changed transition generator

A fixed scene partition contains 2,867 training, 614 development and 615 held-out states. The fitting sets contain 2,048 training episodes and 512 development episodes, shared across conditions and seeds. All marginal values are familiar, but complete held-out scenes are not training scenes. Each of three paired seeds trains both conditions for forty epochs, or 640 updates and 81,920 episode presentations per fit. The optimizer uses learning rate 0.003, batches of 128, global gradient clipping at five and a declared small weight

decay. The checkpoint is selected by development task/stage-balanced accuracy, with lower development log loss and then the earlier epoch resolving ties. Five-epoch development-only pilots are separate and excluded.

All six checkpoints were frozen before generating the 1,024 primary held-out episodes and 1,024 single-entity-edit challenge episodes. The primary generator samples both scenes from the held-out pool. The challenge instead changes both channels of one entity in an unseen reference scene. It is a separately implemented transition generator by the same assistant, not external independent validation. Its edited current scene falls in the training pool on 740 of 1,024 rows, covering 646 distinct training scenes. Reference scenes and complete challenge episodes remain outside training. This overlap is disclosed; the challenge is not evidence that every current state is novel.

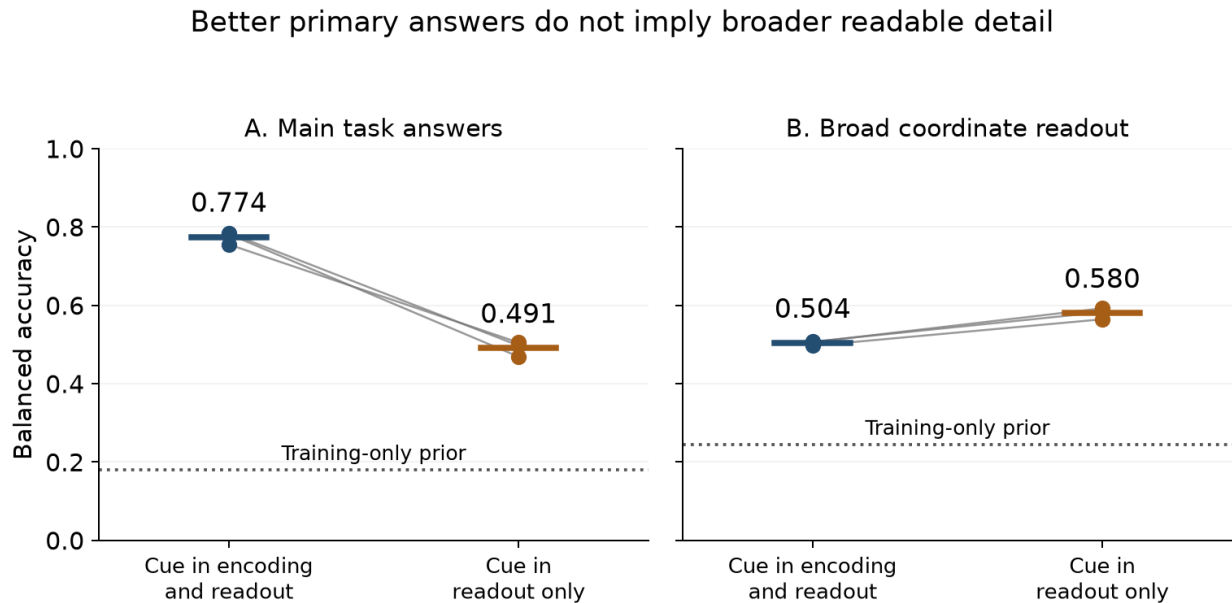
Evaluation includes intact inference, a hidden-state reset at the scene switch, and separately corrupted earlier or current values. Corruption uses fixed across-episode donors while preserving token identities, cues, order and original labels. These interventions assess dependence on correct evidence, not accuracy for relabeled counterfactual worlds. Results preserve every seed and task/stage cell. A training-label-only prior supplies a scene-independent baseline. The 98,304 query-condition rows reuse the same episodes and checkpoints; they are not independent participants or 98,304 independently trained systems.

5.11 Main-task success and broad readable detail rank the models differently

On primary held-out episodes, mean task/stage-balanced accuracy is 0.774 for encoding-plus-readout cue access, versus 0.491 for readout-only cue access; seed ranges are 0.756-0.785 and 0.469-0.505. The corresponding training-label prior scores 0.180. Mean query-weighted negative log likelihood is 0.683 versus 1.488. On the changed-transition challenge, balanced accuracy is 0.781 versus 0.520, with a prior score of 0.174. These are descriptive results for one small training regime and three initializations. A separately optimized or longer-trained late-cue model could perform differently.

I then test broader access without reopening the observations or modifying the recurrent state. Task-stratified affine ridge readouts predict all twelve old/new coordinates from each frozen final hidden vector. Probe labels come only from training episodes; a four-value regularization grid is selected on development episodes. The primary test and challenge are then read with the selected probes. All twelve coordinates include distinctions not needed by a particular current query. Probe performance is therefore not another measurement of the same target or proof that all retained information has been recovered.

The ranking reverses (Figure 7). On the primary test, mean task/coordinate-balanced probe accuracy is 0.504 for encoding-plus-readout cues and 0.580 for readout-only cues; the matched training-coordinate-mode prior is 0.245. Episode-weighted old-coordinate accuracy is 0.376 versus 0.506, and current-coordinate accuracy is 0.633 versus 0.655. Challenge probe accuracy is 0.524 versus 0.616. The less successful main head thus coexists with better access through this broader, separately fitted probe family. This does not prove more total Shannon information, a universal capacity limit or an optimal breadth-versus-detail frontier. It demonstrates why one successful response head cannot identify everything readable from an episode representation.



Three paired initializations; same 1,024 held-out episodes. Points are seeds, not confidence intervals.

Figure 7: Opposite rankings under two declared endpoints on the same held-out episodes. Panel A averages ten main task/stage cells; Panel B averages six current-task strata and twelve probed coordinates. Lines join paired seeds; short bars mark means, not confidence intervals. Probe classifiers are fitted separately from the main heads. The endpoint scales are not interchangeable measures of information or experience.

5.12 Interventions reveal interference and an unresolved temporal task

Figure 8 separates second-query results. In the encoding-cue condition, the delayed-pair answer falls from 0.408 to 0.078 after resetting history and to 0.057 after corrupting earlier values. Corrupting current values leaves this old-pair score approximately intact at 0.416. The readout-only condition shows a similar dependence pattern, from 0.223 intact to 0.063 after reset and 0.055 after earlier corruption. These controlled software results support limited use of prior observations for that answer; neither condition achieves reliable general episodic reconstruction.

Resetting history is not uniformly damaging. In the encoding-cue condition, second-scene fine-coordinate accuracy increases from 0.856 to 0.956 and from 0.792 to 0.911. Broad-coarse accuracy rises from 0.777 to 0.861, and bound-pair accuracy from 0.686 to 0.861. The main ten-cell average also rises slightly after reset. The pattern is consistent with interference from prior state under this learned update/readout combination; it does not imply that deleting human memory is beneficial. The same intervention can remove evidence required by one query and remove interference for another. Averaging them together would conceal that distinction.

The temporal-order task is a specific unresolved failure. Encoding-cue accuracy is only 0.469 intact and is not reduced by reset, scoring 0.480. Its mean log loss, 1.516, is worse than the training-label prior’s 1.116 despite higher accuracy than that prior. On the changed-transition challenge, both learned conditions score approximately 0.34. A current value alone can favor a direction of change, so beating a constant label rule would not demonstrate successful comparison with the reference event. These tests do not establish a robust

learned old/new comparator. The perfect supplied-rule order result in Section 5.6 must not be transferred to this network.

Second-query accuracy (%) separates intervention consequences

	Cue in encoding and readout				Cue in readout only			
	Intact	Reset history	Corrupt old	Corrupt current	Intact	Reset history	Corrupt old	Corrupt current
Fine 0	85.6	95.6	85.8	27.2	47.7	59.7	45.8	25.1
Fine 5	79.2	91.1	84.4	23.2	49.4	55.4	46.3	23.6
Broad coarse	77.7	86.1	81.0	10.2	43.5	58.9	44.4	10.4
Bound pair	68.6	86.1	71.2	7.8	29.0	50.6	30.3	9.1
Past pair	40.8	7.8	5.7	41.6	22.3	6.3	5.5	21.6
Event order	46.9	48.0	46.1	36.6	46.4	40.8	40.5	40.7

ID held-out episodes, three-seed means. Corruption retains original labels; reset is not uniformly harmful.

Figure 8: Task-specific consequences of resetting recurrent history or corrupting observations. Entries are three-seed mean second-query accuracies on primary held-out episodes. Original targets remain fixed under corruption. Improved current-scene scores after reset coexist with impaired delayed-pair recall. The order row does not demonstrate reliable reference-dependent comparison. All first-query probabilities remain identical under later-scene corruption and switch-time reset.

5.13 Reproduction, theorem boundaries and what remains unimplemented

Thirty-six pre-fit checks pass, including 39 finite-difference gradient samples across all thirteen parameter tensors, scene/target checks, no-future-cue checks, readout isolation and exact checkpoint restoration. Eighteen pre-held-out evaluation checks test the metrics, frozen-model identities and probe logic using training/development data only. Fifteen subsequent audit checks recompute stored targets and scores, verify all 480 task/stage cells and preserve exact numeric replay. Prediction and probe-parameter archives reproduce byte for byte. These are assistant-authored internal checks, not independent scientific or implementation review.

The six complete fits took approximately 28.7 seconds in total; held-out inference and probes took about 2.1 seconds on the local machine. Each run used one below-normal-priority process and one numerical-library thread. There was no GPU, model download or background training service. A 48-component float64 hidden vector occupies 384 bytes per episode state; that storage figure is not a count of semantic bits or a metabolic cost. Full parameter, runtime, source, split, checkpoint, prediction and probe receipts accompany the paper.

The representation theorems explain the logical distinction between preserving task-relevant fibers and accessing them with a particular decoder. They do not prove the GRU’s floating-point dynamics, held-out accuracy or probe optimality. In particular, poor linear probe accuracy does not establish a collision or

noninjectivity of the complete hidden representation. Better probe accuracy does not establish subjective richness. The application learns state and answer parameters but not active sensing, an autonomous task-inference process, realistic multimodal content or biological PWD interactions. No published recurrent-attention architecture has yet been reproduced. The next experiment needs to improve and independently test those missing capabilities, not relabel these limited results as their completion.

5.14 Exact measurement checks and a readout counterexample

A separate standard-library diagnostic evaluates the stochastic and readout examples without training or touching the recorded recurrent results. It uses rational probabilities, five binary flip rates, three priors and three bounded loss tables. All 225 postprocessing cases satisfy the risk ordering; output-wise minimization agrees with independent enumeration of every deterministic decoder in those small cases. Another 225 comparisons satisfy the prior-weighted total-variation stability bound. A flip-rate change from one quarter to one eighth attains a risk difference and bound of one eighth under a balanced prior and zero-one loss. These are constructed arithmetic checks, not 450 independently sampled experiments.

The identity and invertibly recoded four-state models both retain all state distinctions. Exhaustive unrestricted readouts achieve accuracy one for the target in both; an integer-coefficient affine search supplies the three-quarter recoded witness. That finite coefficient search is not the proof of the unrestricted affine upper bound; the midpoint argument in Section 4.10 is. The supplied product readout restores accuracy one. By contrast, genuinely dropping the target coordinate leaves optimum one half under the declared uniform model.

Finally, keep an unchanged noiseless binary state and change only the downstream measurement channel. Flip probabilities one eighth and three eighths produce observed optimal accuracies seven eighths and five eighths. The represented state did not change. This construction demonstrates why a recording or access change must not automatically be localized to neural composition or granularity. It does not claim that a real measurement change caused the trained-model results.

All 24 internal checks pass and the evidence payload reproduces identically. The first execution takes approximately 0.079 seconds in one below-normal-priority process; the replay takes approximately 0.077 seconds. No training, download or background process is involved. The test suite supplies consistency checks for known finite constructions, not additional held-out evidence or independent review. Its protocol and exact inputs accompany the manuscript.

5.15 A new ordered-event benchmark removes declared shortcuts

The weak temporal result motivates a new evaluation construction, not replacement of the old result. Each new episode contains two six-coordinate symbolic observations. One focal variable has four circular values. For each fixed background, all four initial values occur with both displacements, one or three steps around the circle; the target identifies the displacement. Reversing the observations reverses the target. The circular encoding is a task device, not a claim about neuronal phase or a substitute for a complete episode.

The prepared dataset has 112 complete background blocks and 896 episodes: 512 training, 128 development and 256 held out. Background combinations are disjoint across splits; focal symbols and the target rule remain familiar. The benchmark construction passes 31 internal checks, including serialization and exact readout enumeration, and reproduces the same 37,779-byte dataset on a second invocation. No network had been fitted at the Draft 3A construction stage. Sections 5.16-5.20 report the subsequent, separately protocolled learned experiment.

For every background and each split, the best accuracy is one half when a decoder receives only the old focal value, only the current value, the unordered fine pair, or both coarse bins obtained by integer division by two. Even access to the entire background cannot improve these restricted scores. Each such view occurs equally often with both labels. The ordered fine pair, by contrast, determines the target exactly. These are exact distributional bounds, not measured learned accuracies, confidence intervals, or additional Lean results.

This construction gives a narrower interpretation to a later successful experiment: the permitted learner would have to use an ordered relation unavailable from either observation alone. It would not identify a specific recurrent algorithm, a PWD mechanism, consciousness, or a general theory of memory. Dataset identifiers, row order and construction variables are excluded from learner inputs; state must be reset between episodes. The benchmark is a small prospective component of the larger episode program, not a new result for the already frozen recurrent models.

5.16 A matched-context comparison joins six episode endpoints

The new study gives both comparators the same observations, supplied roles and answer opportunities. It retains the previous GRU equations [20], a 48-unit recurrent state and a 64-unit nonlinear sixteen-class answer head. A learned context adapter adds 66 weights and eleven biases, producing 14,141 active parameters in each condition. From a one-hot current role, a learned affine map followed by a hyperbolic tangent produces eleven adapter values. The additive model adds one half of this vector to the physical token features. The multiplicative model multiplies those features componentwise by one plus one half of the adapter vector. Both append unchanged role cues before the GRU and provide the current cue to the branching answer head. Zero-initialized adapters make paired initial states and predictions identical.

Both are ordinary context-conditioned recurrent implementations. The additive model can already change its dynamics with the role; it is not a context-blind or fixed-gain strawman. Equal active size and training opportunity do not make function classes, optimization difficulty or operation counts identical. The input adapter is not a faithful replication of a published recurrent-attention architecture, and neither condition is labeled a SAN-specific algorithm. It is a bounded test of two implementations with the same available control information.

More specifically, the additive adapter is redundant with the existing role-input weights at fixed parameters. Write a gate’s physical-input block as P_j , its six role rows as R_j , and its bias as b_j , for $j \in \{z, r, n\}$. A token has physical row vector v and role $r \in \{0, e_1, \dots, e_6\}$. Let $a_0 = \frac{1}{2} \tanh(b_A)$ and $a_g = \frac{1}{2} \tanh(A_{g,:} + b_A)$, where A, b_A are adapter parameters. Removing the adapter and setting

$$b'_j = b_j + a_0 P_j, \quad (R'_j)_{g,:} = (R_j)_{g,:} + (a_g - a_0) P_j \quad (14)$$

preserves every gate preactivation, because $vP_j + rR'_j + b'_j = (v + a(r))P_j + rR_j + b_j$ on that role domain. The recurrent weights and answer head are unchanged. Equal initial states therefore give equal hidden trajectories and predictions by induction, including the same declared reset. The bias term is necessary to cover a zero-role token; merely adding $a_g P_j$ to the role rows only establishes the one-hot case. Arbitrary soft mixtures or multiple simultaneous roles are outside this identity’s stated domain. The actual matched-context data use one-hot roles; zero-role inputs are additional diagnostic fixtures, not new performance results.

A new read-only diagnostic folds each of the three frozen additive models into an ordinary 14,064-parameter core, removing 77 redundant stored parameters from the 14,141-parameter parameterization. Across eight training episodes, three role-token fixtures and both reset settings per model, all 18 comparisons preserve class decisions; maximum probability and hidden-snapshot differences are below 1.5×10^{-15} and $9.6 \times$

10^{-16} , respectively. These floating-point checks accompany the exact algebra; they do not prove equality by sampling alone. Multiplicative adaptation introduces role-by-physical-feature products at the input and does not undergo this particular additive fold. No claim follows that a whole nonlinear GRU cannot emulate those products. An ordinary no-adaptor model trained from scratch remains a required optimization comparator: equivalent frozen functions do not imply identical learning trajectories, regularization or effective capacity merely from matching coefficient counts. The original six fits and their adverse outcomes are unmodified.

Each preserved base episode is expanded across all six final roles. The targets are current coordinate 0; current coordinate 4; the three coarse bins at current coordinates 0, 2 and 5; the fine current pair at coordinates 1 and 3; the fine reference pair at coordinates 0 and 5; and the circular old/new relation at coordinate 4. These require four, four, eight, sixteen, sixteen and two answer classes, respectively. No legal-class mask is supplied. The first role is chosen from the first five tasks by a declared seeded rule constant within each background block, independent of its focal pair and final role. At the first query, current and reference both denote the first observation. The future role is withheld until the second scene.

Each scene presents six tokens in coordinate order. A token contains six coordinate-identity channels, four one-hot value channels, a scene-phase bit and six current-role channels. The model never receives a background ID, target label, displacement label or episode index. Hidden state resets between episodes and carries between scenes; branching answers do not write into it. There are 3,072 training, 768 development and 1,536 held-out two-query episodes, originating from 512, 128 and 256 base pairs in 64, sixteen and 32 background blocks. Replicating a pair across roles creates paired measurements, not six independent situations. These symbolic coordinates are not actual sensory modalities or four-dimensional experienced worlds.

5.17 Fixed-budget learning produces a modest, seed-dependent contrast

Three paired seeds each complete forty epochs, 960 updates and 122,880 training-episode presentations. Both conditions use the same minibatch ordering, learning rate 0.003, batch size 128, gradient clipping at five, declared weight decay and two-query mean negative log likelihood. Checkpoint selection maximizes development mean final-task accuracy, breaking ties by lower final-query log loss and then earlier epoch. First-query performance is reported but does not select the checkpoint. All six selected models, source hashes, training logs and dataset fingerprints are frozen before held-out model inference. The benchmark labels had already been inspected for deterministic balance, not for model selection.

Held-out mean final-task accuracy is 0.607 for additive and 0.621 for multiplicative adaptation; final-query log loss is 1.663 and 1.471. The training-label-only smoothed prior scores 0.224, with log loss 1.934. First-query accuracy is 0.693 and 0.685. Across the three paired seeds, the multiplicative-minus-additive final-score differences are -0.0241, 0.0202 and 0.0443. The mean difference, 0.01345, is not a uniform benefit across initializations.

For the prespecified descriptive analysis, I average the three paired model differences within each held-out background, then resample the 32 background blocks 1,000 times. The resulting 95% percentile interval is [0.00043, 0.02648]. This is conditional on these six frozen models and the constructed background family. It does not include population-level training uncertainty, establish a general architectural advantage or turn 1,536 repeated-role episodes into independent replications. The discordant seed and small mean difference must remain visible alongside the interval.

The experiments in Sections 5.9-5.13 and 5.16-5.20 use different target definitions, generators, ordering and score selection. Their absolute accuracies should not be compared as though one were an improvement on

the other’s test. The new study closes an information-access mismatch in its own comparison. It leaves the earlier ranking reversal, interference pattern and failure unchanged.

5.18 High current-detail accuracy coexists with failed order learning

The current focal-value query is almost perfect: 0.996 additive and 0.990 multiplicative. The current coordinate-0 query scores 0.875 and 0.859; broad coarse composition scores 0.674 and 0.781. The bound current pair remains much weaker, 0.392 and 0.368. These endpoints do not form a single level of rendering resolution.

Both conditions score only 0.501 on the balanced temporal relation. Their order log losses are 0.707 and 0.697, compared with the restricted equal-probability binary predictor’s natural-log loss of approximately 0.693. The separately reported sixteen-class smoothed training prior scores 0.500 with log loss 0.720 because some probability remains on unused classes. Beating that smoothed prior in log loss is not robust temporal competence. A high current-value score and a nearly chance two-event score are compatible: reading a present symbol does not entail comparing it successfully with its predecessor.

The five destructive controls retain the original targets: reset recurrent state at the scene switch; erase earlier value channels; erase current value channels; replace both focal values with equal mixtures within their adjacent coarse bins; or canonically sort the two fine focal values while leaving the background unchanged. Exhaustive equivalence classes of the actual permitted input arrays are exactly balanced for the temporal label in every split. Under reset, the relevant permitted array begins at the second scene because neither hidden history nor an earlier answer is passed forward. Each control therefore has maximum temporal accuracy one half. Both learned conditions score 0.500 under every control. Their intact score is too close to that ceiling to demonstrate dependable exploitation of the ordered information that intact input preserves.

The separate reversal test swaps the entire observations and recomputes both answers. It is a counterfactual task, not a same-target lesion. Temporal accuracy remains 0.501 additive and 0.505 multiplicative. This result does not rescue the failed intact comparator. I do not increase training or change the architecture against this now-inspected held-out set.

History dependence is clearer for the reference-pair query, although accuracy remains limited. Intact scores of 0.206 and 0.225 fall to 0.036 and 0.042 after state reset, and to 0.068 and 0.055 after erasing old values. Erasing current values instead gives 0.250 and 0.281. As in the earlier experiment, history can be useful evidence for one endpoint and interference for another. Reset improves current coordinate-0 accuracy to 0.971 and 0.958, but reduces broad coarse accuracy to 0.594 and 0.654. Neither a uniform improvement nor a uniform impairment describes the intervention (Figure 9).

Reference-pair confidence is also a weakness: intact natural-log losses are 4.773 and 4.257, worse than the smoothed training prior’s 3.012 despite higher accuracy. Selecting the most likely answer and assigning well-calibrated probabilities are different endpoints. The reference result supports limited, history-dependent discrimination, not dependable probabilistic recall.

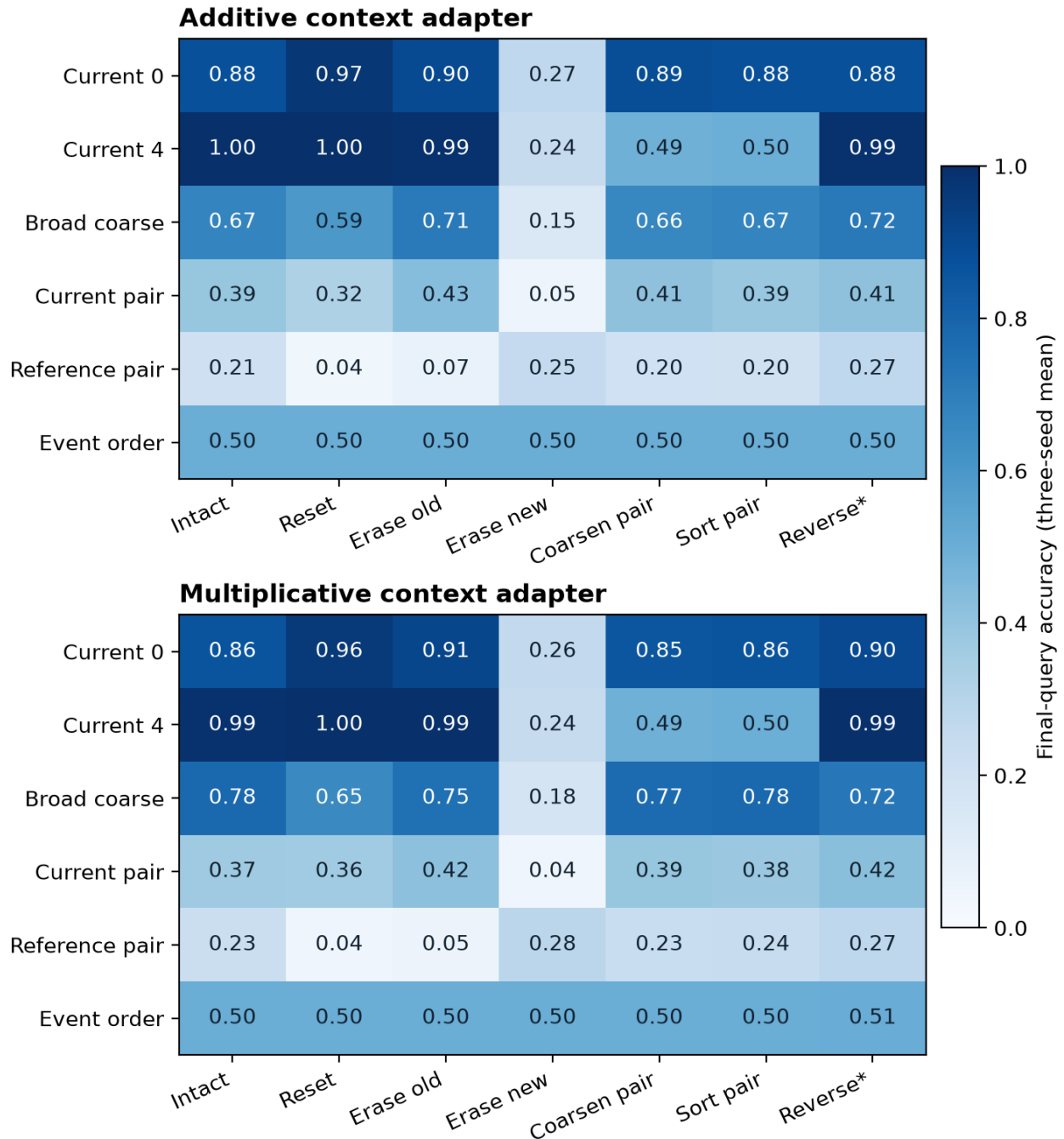


Figure 9: Matched-context experiment: mean final-query accuracy across three paired seeds, with all six tasks and seven input/state conditions retained. Panels use the same scale. The five destructive controls preserve original labels; reversal recomputes labels. The temporal row remains approximately one half even with intact input. Current fine discrimination and reference-pair dependence therefore do not establish successful ordered-episode computation. The cells are repeated measurements of frozen models on the same 32 held-out background blocks, not independent experiments.

5.19 Cross-role readouts expose limited reuse under one probe family

For each frozen model, six affine ridge readouts predict the twelve old/new coordinate values from its final hidden state alone. Each readout is fitted on one final role’s training episodes. Its regularization penalty is selected from 0.01, 0.1, 1 and 10 using mean development coordinate accuracy, with the first grid entry resolving ties. The intercept is unpenalized; there is no hidden-state standardization. Original recurrent and answer parameters remain frozen. I then apply each selected readout to all six destination roles with coordinate identities and targets unchanged. Source-role and destination-role states correspond to the same underlying cases within a split, but no background block crosses training, development and held-out boundaries.

Within-role coordinate accuracy, averaged across roles and seeds, is 0.501 additive and 0.514 multiplicative. Off-diagonal cross-role accuracy falls to 0.306 and 0.314; the training-coordinate-mode prior is 0.242. The decline is present in all six models. Same-role old-coordinate scores are 0.363 and 0.362, whereas current-coordinate scores are 0.639 and 0.665. The aggregate conceals important variation across source roles, destination roles and coordinates, so the full six-by-six matrix and coordinate-level values accompany the result (Figure 10).

This transfer test is not proof that the absent fraction of coordinates has been erased. Section 4.10 supplies a concrete reason why restricted readout performance can change after a lossless recoding. The present study does not identify that exact recoding in its networks. A role-specific affine coordinate map can transfer poorly because its accessible directions change; unprobed nonlinear decoders may recover different content. Conversely, the small above-prior off-diagonal score is not evidence of a universal reusable code. It supports only the recorded amount of reuse under this training procedure and readout family.

The cross-role gap also does not reproduce the earlier cue-access ranking reversal: both main and same-role probe means now modestly favor multiplicative adaptation. These are different comparisons. Treating every mixed result as the same predicted breadth-versus-detail tradeoff would make the proposal unfalsifiable. The next protocol needs prespecified richer readouts, control tasks and learning curves, together with a reason to expect particular contents to share an accessible code. Successful role transfer, successful current action and retained historical information must remain separate endpoints.

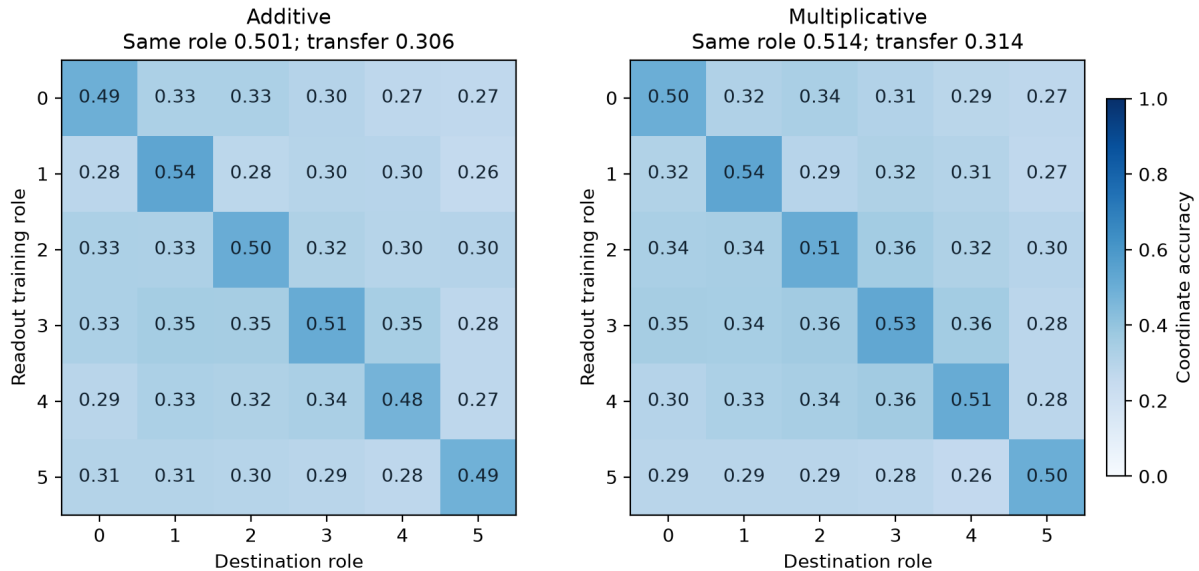


Figure 10: Frozen-state affine coordinate readouts trained on one role and tested on each destination role. Each cell averages twelve coordinate accuracies and three fitted networks. Diagonal cells use matching source and destination roles; off-diagonal cells measure transfer with coordinate targets unchanged. Both conditions have substantially lower off-diagonal means. These are restricted-access measurements, not fractions of total information or conscious richness. The complete per-seed and per-coordinate matrices are retained.

5.20 Verification and the scope of the completed application

Eighty-five pre-fit checks cover target formulas, cue timing, split integrity, matched initialization, numerical derivatives across all fifteen parameter tensors, both adapter conditions, deterministic serialization, role-transfer bookkeeping and the exact temporal input ceilings. After frozen evaluation, 121 internal checks reconstruct all reported task cells, probe outputs from stored coefficients, paired background scores, the bootstrap calculation, source/model hashes and preservation of the previous experiment. Complete probability, state, probe-prediction and coefficient archives reproduce exactly in a second evaluation. Future-only value erasure and switch-time reset leave earlier predictions and states unchanged.

All six fits take approximately 49.0 seconds in total; frozen evaluation and probe selection take about 3.1 seconds, with the replay taking about 3.0 seconds. Every fit stays below its forty-five-second cap. Execution is sequential in a below-normal-priority process with one numerical-library thread, without a GPU or downloads. The 129,024 query-condition rows count six models, seven conditions, 1,536 episodes and two query times; their reuse is explicit. They are not 129,024 independent samples. Checks were authored by the implementing assistant and are not independent scientific review.

The application now supplies an equal-cue learned comparison and a cross-role access measurement on a verified temporal benchmark. It does not supply a reliable learned order comparator, a published architecture replication, an uncued intention learner, a capacity frontier, learned active sensing or biological evidence for PWD assembly. The analytic proofs and twenty existing Lean statements remain unchanged; they do not certify gradient code, floating-point executions or this learned outcome. Those boundaries are part of the result, not qualifications to be silently removed at publication.

5.21 Development-only diagnosis does not rescue the temporal result

After inspecting the matched-context failure, I froze an exploratory diagnostic protocol and used only its original training and development background blocks. No held-out cases were run through a model, and no recurrent parameters were changed. Across the six original models, mean temporal accuracy is already 0.501 on training and 0.514 on development. Thus a failure to generalize to unseen test backgrounds cannot be the sole explanation. Optimization, interference among tasks, retained state and accessible coding remain possible contributors; this experiment does not identify a unique cause.

The diagnostic separates access to the two raw focal values, the earlier hidden state, the final hidden state and both hidden snapshots. Raw values bypass learned memory. Both snapshots add retained history and double the available hidden features; they are not another decoder of the identical final-state evidence. All transformations are fitted using training data alone. Affine ridge penalties are selected from the same four-entry grid. A second ridge readout estimates the two four-class components and then applies a supplied circular relation rule. Invalid decoded pairs abstain and count as wrong. That supplied relation is not a learned end-to-end temporal comparator.

For each view, two 32-unit tanh readouts are trained for 160 full-batch epochs. Selection uses development accuracy, then log loss, then the earliest epoch. Both readout seeds are retained. The raw-pair readouts score 1.000 on training and development, establishing that this small supervised task can be fitted by that procedure. However, all eight focal pairs occur in both splits, so this is neither new-pair generalization nor evidence of learned memory. A raw affine ridge fit scores 0.500; this fitted result is not a proven optimum over all affine classifiers.

Final-state tanh readouts have mean development-selected accuracy 0.565 for additive encoders and 0.548 for multiplicative encoders. Their fixed-epoch-160 accuracies are only 0.527 and 0.504. At that same epoch, training scores are 0.986 and 0.970. Reading both snapshots produces development-selected means of 0.598 and 0.574, but fixed-epoch means of 0.548 and 0.520 despite approximately perfect training fits. These are averages over three frozen encoder seeds and two readout seeds, not six independent datasets.

The negative control assigns four zeros and four ones independently within each background block, with different background assignments in training and development. It supplies no background identifier as a feature. The same final-state tanh procedure attains development-selected control accuracies of 0.577 and 0.563, which return to 0.503 and 0.496 at epoch 160. This is a diagnostic of capacity and selection optimism, not an implementation of the word-type control in [22]. The modest real-label gains cannot be treated as convincing recovered order information when this control reaches similar selected scores. No significance test or estimated selectivity measure is claimed (Figure 11).

The two-snapshot component readout followed by the supplied relation rule reaches training accuracy 0.981 and 0.976 but development accuracy only 0.604 and 0.622. This contrast is also exploratory and changes the access resource as well as the computation. Taken together, the results show learnability of the raw finite rule and extensive fitting of training representations, not a successful rescue of the intended recurrent temporal operation. Further checkpoint or probe selection on the same development set cannot create a new held-out result.

Fifty small nonlinear fits finish in about 4.7 seconds in total, within a roughly 5.5-second single-process diagnostic. Twenty-seven pre-run checks and 315 internal audit checks pass. The audit reconstructs selected and final predictions, component outputs, normalization, selection and metrics from saved parameters. A second execution reproduces numeric results, probabilities and readout parameters exactly. All six recurrent encoders are unchanged. No GPU, download, parallel worker or held-out inference is used. These are implementing-assistant checks, not independent scientific review.

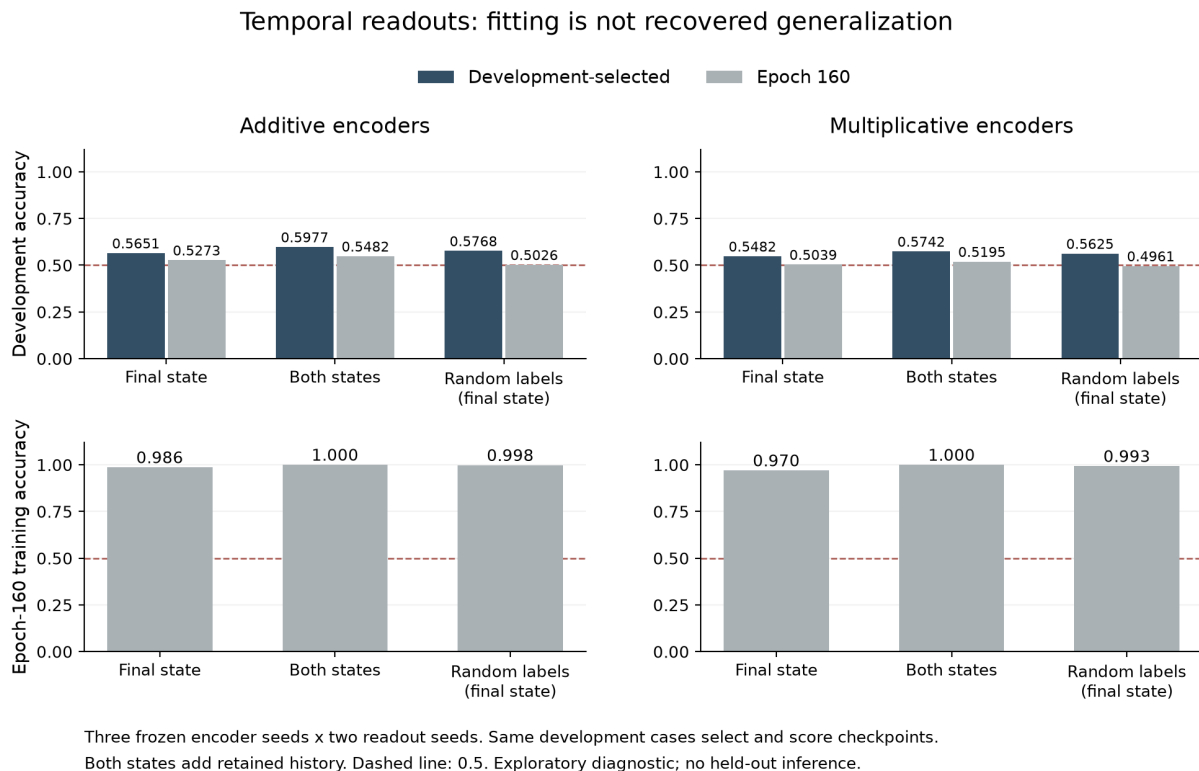


Figure 11: Development-only temporal readouts, averaged across three frozen encoder seeds and two readout seeds. The upper row compares development-selected and fixed-epoch-160 development accuracy; the lower row gives training accuracy at epoch 160. The random-label control uses final-state inputs with new balanced labels per background. It attains similar selected development scores and near-chance fixed-epoch accuracy. Both-state input adds retained history. Selection and scoring reuse development examples, so upper-row selected scores are descriptive, not unbiased generalization estimates. All readouts receive the same epoch budget; raw-pair controls are reported separately in the text.

5.22 Relative order is not the same endpoint as absolute detail

The component readout supplies a sufficient route to the correct relation, not a necessary one. On each case, if both component estimates are correct, applying the correct supplied rule must yield the correct relation. Therefore the relation-error indicator is at most the sum of the two component-error indicators. Taking expectations gives the corresponding error bound without an independence assumption. The pre-run checks enumerate all eight legal true pairs and all sixteen decoded pairs, including abstention. This is an elementary consequence, not a new general theorem or an additional Lean result.

The converse fails. Consider the eight equally likely ordered pairs of four cyclic labels whose differences are one or three positions. A code retaining only that relative difference answers the binary relation perfectly. Each code value is nevertheless compatible with four equally likely earlier values and four equally likely later values. Its best possible accuracy on either absolute value is only one quarter. This two-code construction is checked exactly by the audit. It applies Proposition 1’s distinction-preservation criterion; it does not identify the representation learned by the present networks.

This example matters for the broad composition argument. An action-relevant relationship can remain exact without retaining every constituent at an absolute fine-grained resolution. Conversely, good component access need not mean that the current downstream computation uses the relation correctly. A relative-position code is not automatically a phase-wave differential merely because cyclic labels were used to describe it. The biological bridge still requires the receiver-relative physical operation and a measured consequence. The next experiment must test relation access and component access separately rather than interpreting either as a universal measure of rendering resolution.

5.23 A learned active classifier with encountered episode memory

I next tested a conventional learner that chooses observations and learns responses, rather than supplying the complete current scene or an answer formula during evaluation. The world contains an earlier and a later six-coordinate scene, with four categories per coordinate. A query can request one coarse bin or a fine category. Each first query samples part of the earlier scene; only encountered fields enter the reference record. After an exogenous world change, a second role is requested. A third role uses that same later scene and can reuse its sampled evidence. Temporary assembly is cleared between queries. Missing earlier observations cannot be resampled (Figure 12).

Eight supplied task IDs address current values, coarse context, a current fine relation, an earlier pair, a circular earlier/later relation, or a conditional decision. Complete training examples and labels teach one query tree per task. A node selects one view or a pair of views using empirical label-entropy reduction divided by nominal view cost plus 0.1 per view. A two-view node plans an ordered pair, then attempts requests sequentially. If the first view is unavailable, acquisition stops immediately: the second view is not attempted, and prediction uses the available assembly. Only when all planned views return does the node branch on their values. There is no within-pair replanning based on an available first value, but availability supplies a real early-abort condition. Leaves and partial-evidence predictions use smoothed training counts. The learner is not given the target formulas during deployment, but the tasks, sensor interface, memory structure, supervision and tree grammar are supplied. This is supervised policy learning, not reinforcement learning, optimal planning or a learned neural memory.

Three seeds each receive 128 training background blocks. Four families learn mixed coarse/fine paired queries, mixed single-view queries, fine-only queries or coarse-only queries. Maximum tree depth is eight, minimum child size four and minimum accepted gain 0.02 bits. All 96 task trees are frozen before constructing the 16 development and 16 held-out blocks per seed. Development is descriptive, without checkpoint selection. Each block contains eight balanced focal pairs; symbols and relation rules remain familiar across splits. The first role is fixed within that block and depends on the declared later task, not its answer. The third role is the second plus three modulo eight. These are supplied structured role sequences, not learned intention or arbitrary future-task generalization.

Evaluation adds exactly equivalent attention, complete-record sampling, reference reset, denial of new input, assembly coarsening, binding permutation and inverse restoration. Every query records its source requests, returns, stored fields, assembly, probability vector and costs. The 101,376 records combine three seeds, 128 cases per seed, eight role triples, eleven conditions and three query times. The relevant sampling units are 48 seed-specific held-out backgrounds, not 101,376 independent observations. The full-record control explicitly samples all six fine fields in each scene, paying for both scenes.

Conventional learned query policy with encountered memory

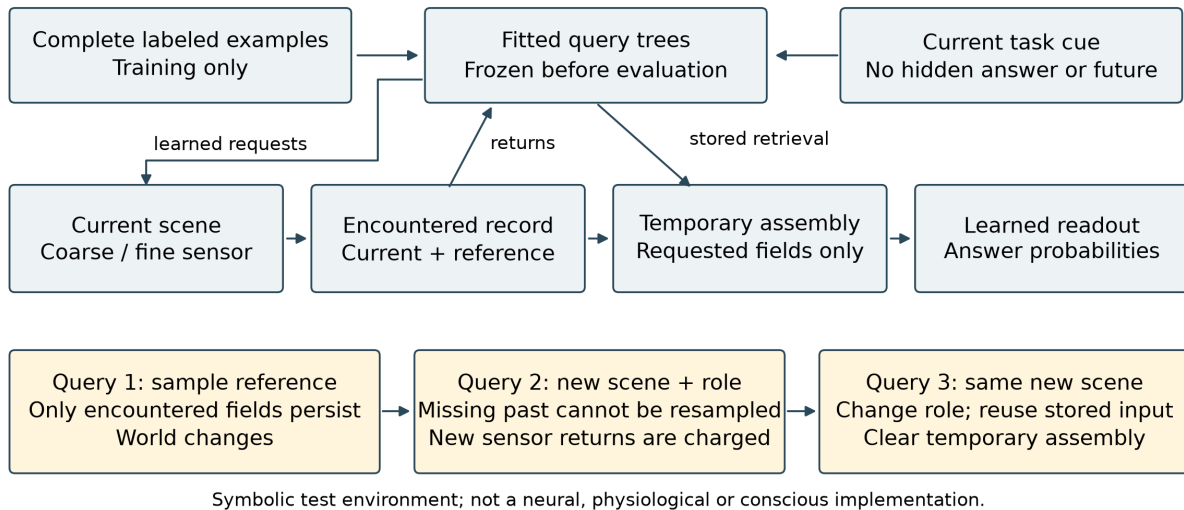


Figure 12: The implemented conventional active classifier. Complete examples are available during supervised training only. At evaluation, the task selects a frozen learned query tree; its requests and returned observations determine later branches. The agent reads only sampled or previously encountered fields. The episode’s reference record persists across the world change, while a temporary query assembly is cleared at each new role. Response probabilities are learned from training counts. The scene transition is imposed by the test, not controlled by the answer. This is an inspectable symbolic application, not a diagram of established SAN physiology.

5.24 Selective acquisition has a cost and a later-query tradeoff

At query two, the mixed learner scores 0.856771, tying fine-only accuracy while requesting 1.960938 versus 2.500000 nominal new source bits. Across all three queries, their acquisition totals are 6.291667 and 7.927083 bits. Fine-only is slightly better at query three, 0.834635 versus 0.832031. Single-view and coarse-only query-two accuracies are 0.833333 and 0.526042. The complete-record control scores 1.000 at both later queries and acquires 24 nominal bits per episode. It is information rich rather than resource matched. Consequently, the result is not uniform superiority of mixed granularity or equal performance against complete memory (Figure 13).

The mixed-minus-single query-two accuracy contrast is 0.0234375 in each seed. A 500-resample paired-background percentile interval is [0.015625, 0.032552]. The mixed-minus-fine source-bit contrast is -0.539063, with interval [-0.559896, -0.515625]. These are conditional intervals over the background blocks of fixed fitted models and authored tasks, not intervals for a human population or a broad distribution of learning problems. Query two and query three contain equal task counts. Nominal source bits count the category width of each completed new sensor acquisition, not planned requests, unavailable views, stored retrievals, Shannon entropy, metabolic energy or total computational cost. A newly acquired coarse value costs one bit and a fine value two; a coarse-to-fine resample is charged two additional bits, not only the incremental distinction. Tree traversal, unavailable attempts, retrieval, storage labels and routing are recorded separately, outside that bit budget. The training heuristic’s extra 0.1 per planned view is also not included in the reported acquisition total.

Reference reset lowers mixed query-two accuracy to 0.820313; denying new input after the world change lowers it to 0.263021. These interventions show use of encountered history and fresh evidence under this interface. They do not establish how an organism internally represents an episode. Complete-record sampling avoids selective-history failures because it acquired more information earlier; a controller that learns to retain evidence for future tasks remains a separate problem.

5.25 Complementary granularity is sufficient only when its history exists

For the balanced circular relation, let the earlier and later values be $x, y \in \{0, 1, 2, 3\}$, restricted to neighbors with $(y - x) \bmod 4$ equal to one or three. A coarse earlier bin $c = \lfloor x/2 \rfloor$ and fine later value y determine the earlier value exactly: $x = 2c + 1 - (y \bmod 2)$. Neighboring values have opposite parity, so this is the unique earlier value in that bin consistent with the later value. The direction then follows. The symmetric fine-earlier/coarse-later code is also sufficient. Two coarse bins, either single value or the unordered pair leave the label balanced. This is an elementary finite witness, exhaustively checked, not an additional Lean theorem or a physical PWD model.

The witness uses three nominal source bits instead of requesting two fine values at four nominal bits. There is no violation of information conservation: the constrained ordered-pair domain already has only eight possibilities. It illustrates complementary granularity, not a universal requirement that every component be fine. The learned tree acquires its query branches and response distributions from training examples; the reconstruction formula above is an explanation, not a supplied evaluation answer rule.

In the mixed learner’s second-query relation task, 72 records retain a coarse earlier focal observation and 72 retain a fine one. Both groups score 1.000. The other 240 records retain neither and score 0.500, yielding an overall 0.687500. At query three, only 72 relation records retain the required earlier value, while 312 are at chance. These are repeated cases under role sequences, not independent participants. A useful relative relation therefore depends on both complementary access and what the earlier role caused the system to encounter. This new tree result does not repair or replace the preserved near-chance recurrent neural models.

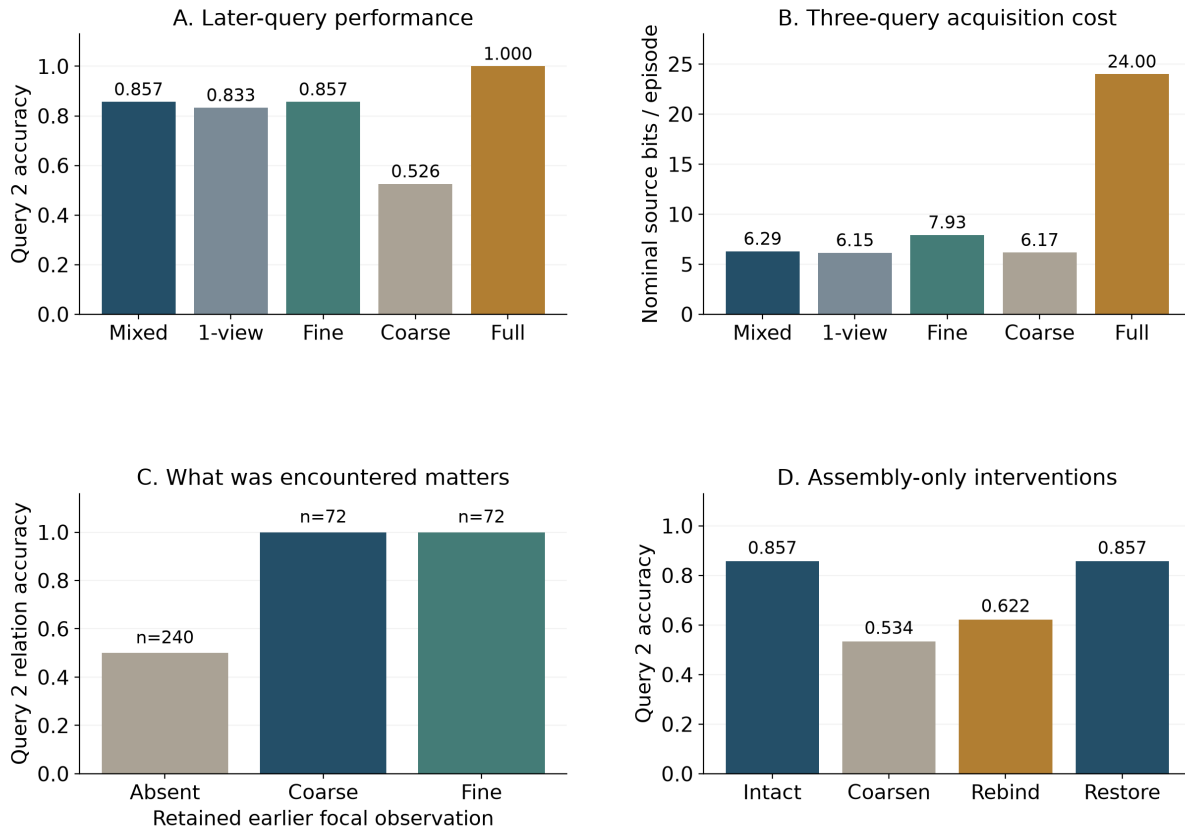
The acquisition policy matters to this cost interpretation. In all 240 unavailable-history records at query two, the mixed root attempts only the earlier coarse view (feature 8), records its unavailability, and never requests the planned current fine view (feature 21). Those records acquire zero new source bits. The reported cost advantage is therefore conditional on the implemented availability-triggered abort as well as the chosen view widths; it is not the cost of an unconditional two-view policy. Six new fixtures across the three frozen mixed trees check both branches of this contract: missing history yields one unavailable request and no sensor call; available coarse history yields one retrieval followed by a two-bit current acquisition. A direct readback of the unchanged historical archive reproduces the 240/384 count and the reported 0.856771 accuracy and 1.960938/2.500000 mixed/fine costs. This correction changes the explanation, not the policy, weights or frozen results. The revision addendum accompanying the original protocol records the same contract.

5.26 State-contingent requests and assembly-specific interventions

The conditional task supplies a simple learned sampling dependency. When the encountered coarse gate selects the lower branch, the mixed learner requests one coarse bit and answers correctly. For the upper branch it requests an additional fine value, totaling three bits, and also answers correctly. The second-query groups contain 224 and 160 repeated records. This is a change in requested granularity and composition conditioned on observed symbolic state, not evidence about cannabis, fatigue or another physiological condition.

After acquisition, coarsening the temporary assembly changes query-two accuracy to 0.533854. Permuting two bindings within each time slice changes it to 0.622396. The source actions and stored evidence remain exactly unchanged in these interventions. Inverse restoration and the equivalent attention interpretation reproduce the original probabilities, actions, cache, assembly and costs exactly. Thus the program exposes distinct manipulable variables, while retaining an ordinary explanation and an algebraically expected rescue. It does not identify a distinctive neural circuit or outperform the published active-acquisition approaches in Section 2.6.

Forty-one current pre-fit checks pass. An initial tautological role test was corrected before fitting, with its prior receipt preserved. Fifty-two post-result internal checks reconstruct every held-out query's observed actions, retained cache, temporary assembly, probabilities and costs, and recompute the training counts of 2,676 learned tree nodes. Held-out predictions, cases and bounded success/failure traces replay byte for byte. Training takes about 1.6 seconds, each evaluation about four seconds and the audit about ten seconds in a single below-normal-priority CPU process. These are implementing-assistant checks, not independent review. There are no new biological data or formal compiler results.



Three fitted seeds; 16 held-out background blocks per seed. Repeated queries are paired.
 Full access costs more and performs better. Ordinary-attention interpretation ties Mixed exactly.
 Counts in C are repeated queries, not independent participants. Costs exclude routing and computation.

Figure 13: Recorded active-classifier results. A: task-balanced query-two accuracy. B: nominal source acquisition across all three queries. Mixed and fine-only tie at query two; full access costs more and performs better. C: mixed-policy relation accuracy grouped by the earlier focal value actually retained; group counts are repeated query records. D: assembly-only coarsening and rebinding leave source actions and cache unchanged, while inverse restoration reproduces the original result. Equivalent attention also ties exactly. Means use three fitted seeds and 16 held-out background blocks per seed; the text gives paired conditional intervals and the slightly better fine-only third-query result. These results test an authored symbolic family, not experienced multimodal episodes.

6 A research program capable of distinguishing the proposal

6.1 Same environment, different reconstructed episode

A first behavioral study should hold a controlled multisensory environment as comparable as possible while changing the required role. Trials should probe fine local distinctions, which elements were represented, which relations were bound, the order and extent of remembered events, and predicted consequences.

Delayed probes can test what becomes accessible later, without assuming that inaccessibility means permanent erasure.

Role differences must not be inferred solely from questionnaire descriptions. Relevant endpoints include discrimination thresholds, relation-specific errors, reconstruction accuracy, calibrated confidence, and action selection. Reports of felt richness should be analyzed separately. A role manipulation that changes only response policy, with no change in the independently assessed representation, would not establish composition-and-granularity reconfiguration. Proposition 4 shows why even perfect primary performance through task switches is insufficient. Independent assessment should therefore separate pre-probe state measurements from changes caused by the probe. A failed auxiliary readout may reflect measurement or decoder limits, and a successful delayed probe may reflect newly recruited memory.

6.2 Separate external sampling from internal reorganization

A free-sampling condition should permit eye, head, and other safe information-gathering movements. A matched or replayed-sampling condition should equate available evidence. Otherwise, better discrimination after a closer look could be attributed to improved sensory input without demonstrating an internal representational change.

A useful design crosses composition demand with granularity demand: which modality or relation matters, and how finely its alternatives must be separated. Temporal-span demand should be varied independently where possible. Decoder analyses must use held-out data, preserve task-label separation, and compare models of gain, uncertainty, attention, context-dependent recurrent computation, and joint reconfiguration. All models should receive equivalent task information and a fair accounting of control costs.

The V1 integration/segregation result [28] suggests a sharper population-level test, not a substitute for the full episode experiment. Estimate intrinsic subspaces from independent baseline data, freeze the axes, then measure task-conditioned movement and stimulus projections on held-out awake trials with input, movement and state controls. Score separately (i) useful interaction between modalities or body and stimulus, and (ii) preservation of distinctions that would otherwise interfere. A role cue with the same physical scene should not be called composition tuning merely because a total-activity or decoder score changes: test whether the *relations assigned to these components* and their downstream use change beyond a fixed-geometry, task-conditioned readout. A null result on this V1 slice would limit that slice; a positive result would not by itself establish multimodal SAN rendering, PWD, NAPOT, or experience. The source’s public data archive is large and was not downloaded or reanalyzed for this paper.

6.3 From a symbolic fixture to a learned, closed-loop application

The recurrent comparison now learns representation and answer parameters beyond the earlier acquired symbol dictionary and macro-action selection. Its limits define the next application: integrate learned active acquisition with recurrent episodic state, improve temporal dependence and calibration, and test realistic multimodal relationships without a complete task-head grammar or compulsory full familiarization. The separately coded transition challenge is not an independently authored validation generator. Training and evaluation families, ordinary attention and strong context-conditioned recurrent comparators require further development. Ablations should remove relation assignment, variable granularity, temporal maintenance or returned feedback one at a time while preserving access to the same source information.

Selective perturbations and rescues should discriminate failure to acquire content, failure to preserve a distinction, and failure to route it to the appropriate readout. Outcomes should include held-out task

performance, reconstruction errors, switching recovery, calibration, and total control-plus-representation cost. A tie remains informative. If established models explain every result equally well, the framework may remain a useful synthesis without a new mechanistic advantage.

The comparison must also include a full-information encoder with a task-conditioned readout, so a learned decoder is not mistaken for representational reconfiguration. Equal state boundaries, opportunities to resample, and declared memory access are essential. The newer recurrent gating comparator [18] should retain its binding and memory operations. Task switches with reliable cues must be analyzed separately from task inference based on consequences.

The present implementation now separates the four inspectable objects specified in the earlier draft: acquired associations, encountered episodic memory, temporary assembly and readout. Selected sampling actions change the next observation; unseen world values are not supplied as prior state. Its episode split and frozen readout probes are implemented. These boundaries should remain explicit in a richer successor, where task semantics, relationships and retrieval policies can themselves be learned. A supplied correct role label must still be reported as such, not described as inferred intention. A probe that reopens memory is a new state transition, not the same frozen-state measurement.

6.4 Requirements for a biological PWD claim

A biological test must specify a source event, a receiver-relative reference, receiver state, timing, and a declared consequence. Merely observing more functional connectivity or a changed oscillatory spectrum is insufficient to identify the proposed operation. Causal perturbation should alter the predicted route or timing while suitable controls preserve alternative explanations; a targeted rescue should restore the expected relation when feasible.

The anatomical implementation and the conscious-status claim require separate evidence. Access to an episodic detail, accurate action, neural interaction, and conscious experience cannot be equated by definition. The present program concerns testable representational organization first. It proposes possible relevance to experience without treating success on a symbolic task as its measurement.

6.5 Test the transition while preserving the whole episode

A more diagnostic prospective experiment would present a continuing, controlled audiovisual episode with spatial, social, bodily-action, and temporal relations. A cue would change the required role without replacing the entire scene. Independent manipulations would vary which relation matters, how finely alternatives differ, and whether immediate or longer-range consequences are required. Safe sampling opportunities would be either free or matched across conditions.

The primary question would be how the measured configuration changes before and after the cue, not whether one role produces more total activation. For example, does an action-relevant local distinction improve while broad contextual relations remain available, become less accessible, or are rebound to different consequences? Does returning to the earlier role recover a prior organization, require renewed sampling, or construct a different one? Immediate discrimination, relation errors, event-order judgments, confidence, and later reconstruction should be analyzed separately.

This is a proposed study, not a report of participants or a claim that present measurements can uniquely recover K_t . The readout-only countermodel, external-sampling explanation, context-conditioned recurrent models, and learning-related changes all remain live alternatives. Additional modalities and bodily or

pharmacological conditions would require their own validated manipulations and appropriate ethical review. An arbitrary parameter in a simulation must not be labeled cannabis, meditation, or fatigue.

6.6 Measure access without confusing it with assembly

The proposed four-part configuration must be connected to a declared observation chain: episode conditions affect neural state; a recording samples that state; a probe transforms the recording; and a decision uses the result. A change at any of these stages can change measured performance. For example, across the arena roles, a less successful fine-detail probe could reflect coarser internally available detail, an unchanged detail encoded differently, fewer relevant recorded cells, or a receiver that no longer accesses it efficiently. Calling every such outcome granularity tuning would make the proposal uninformative.

A prospective study should therefore fix the comparison’s task distribution and sampling boundary, measure primary and auxiliary endpoints separately, and compare a prespecified sequence of readout families with matched training opportunities. An affine probe and a flexible probe answer different questions; neither is automatically the ground truth. Probe training must be separated from assessment, and capacity and control-task checks must be designed before examining the final evaluation set [22]. The development-only diagnostic in Section 5.21 adds a bounded capacity/control check after the earlier test was inspected. It is not a retrospective prespecified test-set analysis; the broader prospective controls remain necessary.

Composition concerns which contents become effective together; binding concerns their assignments; granularity concerns distinctions relevant to a declared endpoint; temporal extent concerns which event relations remain available. Evidence for these need not produce the same ranking. A controlled loss of a particular relation with preserved constituent access can motivate a binding account, but an invertible recoding or inadequate relational decoder remains an alternative. A shorter effective episode requires old-event tests that cannot be answered from current evidence alone. A finer odor distinction requires control of sampling and the link to remembered or anticipated consequences, not simply greater recorded activity. These are ways to test my integrated-episode argument without reducing it to vision or assuming that more activity implies more information.

The stochastic model gives a sufficient route to rigorous comparison when channels can be identified or bounded. It does not promise that partial brain recordings permit that identification. Where the evidence only supports a decoder-relative change, that is the appropriate conclusion. Demonstrating a common scalar ordering, a neural control operator, or conscious experience would require further assumptions and evidence.

6.7 Separate inferred role, reused code and retained history

The matched-context study in Sections 5.16-5.20 now gives two comparators equivalent task cues, observations and evaluation opportunities, with active as well as allocated parameter counts disclosed. This satisfies a bounded control requirement, not the complete comparison program. Retain a cue-deprived encoder as an explicit information-access ablation, but do not let that ablation alone support an architecture claim. A context-conditioned recurrent model and a learned modulation model must retain their actual capabilities. Neither is represented adequately by a fixed amplification factor.

After learning is frozen, train a decoder on one role and assess it on another while holding the target relation and scoring rule fixed. Pair that transfer test with a decoder trained within the destination role. Successful transfer would support a shared readout under the declared family. Failed transfer with successful destination training would support changed access or coding, not necessarily missing content. If both fail, test prespecified richer readouts and learning curves before attributing the failure to representation. Match

motor/report mappings or counterbalance them so the probe cannot merely decode the response habit. Probe training and its assessment must not reuse the same episodes.

An uncued-role condition is a separate experiment. It requires the agent to infer the operative task from permitted history, uncertainty and feedback, rather than silently receiving the answer through a task token. Role belief, content access and action success should be measured independently. These are proposed artificial controls, not a reproduction of the animal study or a completed latent-context learner.

The ordered-event component in Section 5.15 supplies a new test boundary for temporal dependence. It can be combined with current-detail and broad-context queries, retaining equal context information across models. However, adding role queries or state manipulations can break its balance unless the complete blocks and conditional label checks are retained. Analysis must be clustered by background block; the 896 enumerated episodes are not 896 independent biological observations. A new learned result must retain failures and report uncertainty appropriate to the sampling and initialization units.

The broader biological question remains an integrated episode: what relations become jointly consequential for this role, at this moment, with this history? The component tests identify several ways a measurement can answer a narrower question than that one. A successful joint account must predict more than task switching, decoder transfer, or a two-event score separately, and it must remain useful when the existing recurrent and attention explanations perform just as well.

6.8 One operational joint-configuration test: a remembered commitment guides a later action

The following is a **prospective artificial experiment**, not an executed successor study. It supplies one finite, inspectable instance of the full question without treating that instance as the whole SAN theory. The scene is a simulated arena with two named actors, two entrances, a self-location, visible entrance states and an auditory commitment made earlier by one actor. The commitment links a four-valued spoken token to an intended entrance and time. Later, a role cue asks for either a broad management reconstruction or a specific route decision that depends jointly on who made the commitment, its exact token, the current visible entrance and the agent’s present location. Earlier actor-to-voice assignment and current left/right location are counterbalanced, so neither the words alone nor the current image alone determines the action. The inputs are typed event records standing in for audiovisual and bodily observations; this first implementation is a relational-memory test, not a trained speech/vision system or measured experience.

Encounter, learning and sampling boundary. Every model receives the same four-event observation stream, including the same initial commitment. Durable association parameters are learned only on training episodes and then frozen. An encountered episodic archive stores only supplied observations; an assembly controller may read that archive at a charged interface, but neither it nor its decoder may read latent targets. At the role switch, external input is clamped to the same recorded event and no new sensory sample is permitted before the primary snapshot. The controller receives the role cue, prior assembly and available archive and must produce its next assembly within four update steps. Its chosen queries, slots and associations must be learned from development-limited training, not supplied from the correct answer. The answer head is frozen and receives the resulting assembly and role cue; its computation cannot silently reopen memory. A separate restoration condition can reopen the archive and must be counted as a new transition. State and archive reset between episodes; all within-episode history is preserved unless explicitly perturbed.

Declared state and measurements. The candidate implementation has a typed slot-and-relation graph with a fixed 64-bit payload budget; role/event tags, graph pointers, queries and controller work are charged

separately rather than hidden as free information. It emits an inspectable configuration from the actual records present at each snapshot, not a configuration inferred solely from final correctness:

- C is the set of source variables whose payloads are present in the assembly, with event and modality labels. A stored but unread archive field is not counted as participating content.
- B is the graph of actor, speaker, entrance, self-location and event-role assignments between those payloads. Its exact-match score uses separately recorded experimental identities, never identities supplied to the policy as answers.
- Π is the equality partition of assembled codes on a finite counterbalanced panel. For the critical spoken token, test all four values with the other declared variables held fixed; distinguish exact four-way retention from the two-bin coarsening $\lfloor v/2 \rfloor$. A payload’s absence is reported as absence, not mislabeled as a coarsened measurement on the same participating set.
- H is the set of represented event lags and their order edges. The primary contrast is access to the earlier commitment versus a current-only assembly, with current observation and archive held constant. This measures active temporal availability rather than how much history exists somewhere in storage.

These coordinates are dependent. Removing an old auditory payload changes C , removes its binding edges from B , may collapse Π , and may shorten H . Therefore the analysis first reports that joint transition, then compares granularity only on matched present payloads, binding only when the needed constituents remain available, and temporal effects only at fixed current evidence. It never claims four independent biological knobs merely because four software fields can be logged.

Identifying interventions. Save the post-cue assembly once, before the answer, and fork read-only evaluation copies. In one copy swap the two speaker-to-commitment edges while preserving payloads, event tags, granularity and archive. In another coarsen only the critical token within its predeclared pair while preserving which token belongs to which actor. In a third remove the earlier-event access while preserving all current payloads. A fourth applies the exact inverse binding permutation. Inspect the graph and constituent readouts to verify what each operation actually changed. Any failure of those manipulation checks is a failed test implementation, not a result about the hypothesis. In particular, deleting old content is a joint $C/B/H$ intervention and must not be advertised as an isolated biological manipulation of time.

Joint prediction. The finite candidate is committed to a specific conjunction: swapping the remembered speaker assignment changes the predicted route only when the decisive old token is present at sufficient granularity and is linked to the current actor/entrance relation. Select balanced episode pairs in which the correct route reverses under that swap while current visual input, self-location and individual token values remain fixed. In intact long-history/fine-token trials, the intended assembly should follow the swapped relation; inverse restoration should recover its original answer. In matched coarse-token or current-only trials, the exact route is balanced and the missing distinction cannot be restored by naming the role. This is a prediction about the specified finite representation and readout, not a claim that all real memories use these discrete slots.

Let $E_B(p, h)$ be the fraction of balanced pairs for which the answer changes in the specified direction after the binding swap, where p indexes fine versus coarse token retention and h indexes earlier-history availability. The primary interaction is

$$J = E_B(\text{fine}, \text{earlier}) - E_B(\text{coarse}, \text{earlier}) - E_B(\text{fine}, \text{current}) + E_B(\text{coarse}, \text{current}). \quad (15)$$

Report the four constituent terms, correct route/reconstruction scores, constituent token access, exact configuration changes and all charges alongside J ; a positive interaction alone is not identification. For this finite candidate, the prospective engineering success target is intact route accuracy at least 0.90, $E_B(\text{fine}, \text{earlier}) \geq 0.80$, the absolute value of each unavailable-distinction term at most 0.10, and inverse-

restoration probability differences at most 10^{-10} when the representation and decoder are deterministic. These thresholds are chosen software targets, not inferred physiological constants. Exact conditional label balance and the intervention checks must pass before interpreting those scores. A decoder may exploit residual differences in an incorrectly constructed coarse condition; that would invalidate the manipulation rather than demonstrate recovered information.

Countermodels and held-out boundary. Compare the learned assembly candidate with an ordinary context-conditioned recurrent model with the same input/archive/query interface, observation and payload budget, training allowance and answer opportunities. Also include a role-conditioned decoder of one unchanged encoded episode; it must receive the same initial observations, use the same declared access boundary and have its storage/readout costs disclosed. A complete-record unconstrained decoder is an additional information-rich upper control, not a cost-matched competitor. No candidate is denied the ordinary relational and recurrent capabilities supplied to the proposed model. Effective parameterization, rather than stored coefficient counts alone, is reported in light of Section 5.16. Build 64 development background blocks for model selection; after source/model hashes and analysis choices are frozen, generate 256 new background blocks containing the full within-block speaker, token and role counterbalances. Five fixed fitting seeds (2027, 2039, 2053, 2063, 2081) are proposed. Cluster uncertainty by background and show individual seeds; the many paired interventions are not independent subjects. No inspected legacy test case becomes fresh evidence in this new protocol.

Failure of the declared direction or threshold with verified access, intervention and successful optimization controls rejects this finite candidate’s claimed realization, not every SAN implementation. A failure to train it is a construction failure rather than a biological falsification. If an unchanged-episode or ordinary recurrent countermodel matches the complete measured signature, the experiment has not identified a distinctive mechanism or advantage; it remains a useful diagnostic decomposition. Conversely, an advantage on this task would require reporting information, storage, computation and optimization differences before attributing it to the joint control. Transfer to biological recordings would still require independently measured receiving state and causal pathways. None of the targets above has been tested in the present paper. The contribution now delivered is a synthesis and diagnostic benchmark framework with an operational joint test specified, not an identified biological joint-control law.

6.9 An executed measurement diagnostic: the same joint signature without a changed stored episode

Before undertaking the learned study, I tested its measurement logic in a separately declared finite diagnostic. The hand-specified arena supplies two actor/voice assignments, complementary within-bin four-valued auditory tokens, two current entrance states and self-location. A queried actor’s earlier token, actor binding and present situation jointly determine the route. This is not merely a sharper image: retaining the wrong speaker relation changes what the situation calls for even while its objects and tokens remain present.

The 128 counterbalanced base episodes produce 512 fine/coarse by earlier/current snapshots. The constructor explicitly records participating payload identities C , speaker edges B , token partition Π and represented lags H . Current vision and proprioception share lag zero; two earlier commitments have lags -3 and -2. Four typed records therefore have three distinct times in this narrower diagnostic. The constructor and answer rule are programmed, not learned. Neither the management endpoint, a trained query policy, the proposed 64-bit implementation, five fitting seeds nor a new held-out background split is delivered by this test.

The intact fine/earlier answers are exact. Binding swaps change only the speaker edges in saved evaluation copies; the supplied content, granularity and time labels remain unchanged. Inverse restoration recovers the original output exactly. Coarse/current representations have balanced exact-route labels across all 80

distinct incomplete-observation groups, so no permitted decoder can recover the missing low-bit distinction from those groups. The four measured E_B terms are 1, 0, 0 and 0; hence $J = 1$. All cases and checks remain available, including byte-identical replay and 6,352 contract assertions. This verifies the hand-constructed measurement example, not a learned achievement of Section 6.8’s success thresholds.

The crucial control has an unchanged complete-record memory. Instead of changing that memory’s encoded assembly, it applies the same binding/access/coarsening transformation only in the answer computation. Its stored record hash remains unchanged and its full probability/intervention signature, including $J = 1$, equals the assembly-changing construction. The record is information-rich; this is not a capacity-matched or equal-work benchmark victory.

The underlying identity is simple. If an assembly constructor under intervention q produces $V_q(e)$ from episode e , and the answer rule returns $d(V_q(e), r)$ for role r , a fixed-record implementation can compute that same composition at readout. Its behavior agrees for every episode in their common domain. This is a factorization identity, not a claim that storage, effort or physical intervention targets are equal. Directly observed state changes distinguish these two software constructions; their actions alone do not.

This result makes the learned experiment more discriminating: the positive interaction must accompany independently verified constituent and assembly measurements, the correct access boundary, and a fair ordinary recurrent alternative. A named “binding intervention” must not silently alter the decoder in one model while altering retained structure in another and then be called physically matched. The primary theory remains the task/state-dependent assembly of a multimodal remembered and prospective situation. This diagnostic identifies a false-positive interpretation of one proposed score; it neither establishes nor rules out the biological implementation.

7 Limitations and claim boundaries

This is a source-, application- and formally supplemented **theory-and-methods preprint**, not a validated general theory of state-dependent cognition or an accepted completion of the wider empirical research program. The V1 primary comparison supplies an integration/segregation test design, not new application data or a demonstration of the complete SAN conjunction. All five selected neighboring public manuscripts have been read in full extracted text, but the complete corpus and historical genealogy remain open. Method-level review preserves the substantial recurrent-gating alternative; reproducing its published architecture remains open. The propositions and bounded Lean results establish consequences of declared encodings; they neither identify a unique neural implementation nor certify the complete SAN architecture. The first two diagnostics use exact finite models; the third learns a symbolic decoder and macro-action policy under supplied task semantics. Equivalent attention ties that selection. The fourth trains recurrent representations and answers under unequal encoder-cue access; its main-task advantage reverses under broader probes and does not generalize to reliable temporal-order computation. Restricted-readout failures remain distinct from what another decoder can recover.

Composition, binding, granularity, and temporal extent may be difficult to identify independently from partial neural observations. A measured geometry can change for several reasons. A failed probe can reflect retrieval, decision, motor, or measurement limitations rather than absent content. Experimental conclusions must survive such alternatives before they are assigned to the proposed configuration coordinates.

Historical sources establish the content and development of an argument only to the extent their wording, dates, and custody are verified. They do not independently validate biological mechanisms or establish

priority over every relevant scientific program. Later terminology is not projected backward into earlier notes.

The strongest version of the hypothesis is therefore conditional: if state- and task-dependent changes in episode composition and granularity can be measured, causally separated, and predicted beyond existing alternatives, the joint framework may explain a useful layer of neural rendering control. Until then, it is a structured research proposal with transparent mathematical and computational boundaries.

The new stochastic extension is finite and conditional. It compares known channels under matched priors, targets and bounded losses; it neither estimates neural observation channels nor proves that the trained conditions form a postprocessing order. Its exact examples are explanatory counterexamples, not biological data or new learned performance. The unrestricted-versus-affine distinction adds a concrete reason why the recurrent probe ranking cannot by itself identify altered information retention.

The newer matched-context study is small, synthetic and supplied-role. The context-adaptor contrast is seed-dependent, and near-chance temporal results remain a genuine failure of this implementation to realize the desired ordered-event operation under its fitting budget. They are not evidence against every SAN implementation or evidence that the benchmark is impossible; an ordered fine lookup already attains perfect accuracy in its declared finite model. They are also not grounds to declare the intended learned operation complete. Repeated fitting to this test set would weaken the next evidence claim. A genuinely new training strategy requires development-only exploration, a separately frozen test boundary and retained adverse results.

The diagnostic in Sections 5.21-5.22 does not close this gap: it changes readout capacity and, in one condition, history access, while development selection produces optimistic scores even for random labels. It establishes neither a unique cause of failure nor successful order generalization. All twelve frozen recurrent models across both earlier studies remain part of the package.

The active classifier in Sections 5.23-5.26 learns queries and responses, but has supplied symbolic memory and a supervised finite task family. It does not learn recurrent neural state, sensory transduction, long-horizon retention or an inferred role. Observations are exact categories rather than noisy measurements; no unfamiliar relation rule or independently designed environment has been tested. The mixture rule for incomplete tree evidence is a particular training-count estimator, not a certified Bayesian posterior. Assembly perturbations are software operations after acquisition. The complete-record advantage and the conventional-attention tie prevent this study from being interpreted as a universal resource or architectural advantage.

The planned stronger empirical test-one learner that acquires and retains multimodal episode evidence across changing roles while its recurrent state and sampling policy are both learned-has **not** been completed. The present manuscript is final only as a testable decomposition, formal boundary and transparent critical application package. It must not be described as a demonstrated neural rendering-resolution controller. EDDI, online feature-acquisition trees, nongreedy subset acquisition and recurrent gating remain serious alternatives; their targeted methods were read, not reimplemented or benchmarked here. The next study must freeze a new evaluation family after model selection, use matched information and cost, and retain the adverse fine-only, full-record, attention-tie and temporal-order results.

8 Conclusion

The question is not only how small a difference a brain can detect. It is how much of a learned organization becomes available now, which relationships assemble it, which distinctions survive, and what situation thereby becomes available for reconstruction or action. Composition and granularity name different parts

of this question. Temporal extent, binding, bodily state, and returned evidence prevent the theory from collapsing into a visual sharpness metaphor.

The applications demonstrate why task-appropriate representation matters and why conventional attention must be taken seriously. The symbolic controller separates stored evidence, temporary assembly and reacquisition; the first recurrent experiment learns its own state and answers, but better main-task performance coexists with worse broad probe access and weak temporal comparison. The new equal-cue comparison adds limited cross-role reuse and near-chance order learning despite strong current-detail performance. These differences make the proposed measurement distinctions concrete without establishing a new neural algorithm. The mathematical model makes sufficiency and incomparability precise without converting them into a proof of consciousness. Strong published comparators, integrated learned sensing and discriminating biological evidence remain necessary.

The new active classifier makes one further boundary concrete: an observation policy can learn to combine coarse and fine evidence while still failing later tasks because it did not encounter the required past. That tradeoff connects composition, granularity and temporal availability without reducing an episode to visual sharpness. The next implementation should join learned recurrent state with learned acquisition and retention, compared fairly with existing methods. Successful behavior alone remains insufficient to identify the proposed biological organization or conscious experience.

9 Data, code, and assistance

This local review package includes two preserved finite diagnostics, the symbolic learned episode application, twelve frozen recurrent models across two experiments, their protocols and source, training and development logs, held-out predictions, interventions, restricted probes and exact replay checks. It also includes the earlier event records and completion analysis, analytic proofs, Lean source and compiler/axiom receipts, source intake manifest, research plan and versioned review records. It contains no newly collected human or animal data. Public archiving, licensing, independent review and author approval of a release remain pending. AI assistance was used in source routing, literature checking, drafting, code preparation and analysis; this is not independent scientific review. Historical AI-assisted interpretation is distinguished from original author statements in the accompanying provenance records.

The Draft 3 package additionally includes the stochastic measurement protocol, exact-rational diagnostic, repeatable result payload, hand proofs and claim-to-evidence map. Earlier learned parameters, evaluation files and proof receipts remain unchanged. The numbering marks a substantive working revision; it does not close unfinished genealogy, comparator, temporal-learning, independent-review or release requirements.

Draft 3A added the bounded scientific method review, the ordered-event construction protocol, a small serialized dataset and two identical construction-check outcomes. At that stage the dataset had not produced a learned result. Draft 4 adds the matched-context protocol, six further frozen models, complete held-out predictions, cross-role coefficients and predictions, a replay and an internal audit. The earlier supplied-cue experiment remains unchanged, including its weak temporal performance. The new failed temporal result is also retained. Published-model replication, uncued task inference, integrated active sensing and independent review remain open.

Draft 4A adds the exploratory temporal-access protocol, readout sources, learning curves, fitted readout parameters, exact replay, arithmetic audit and a selection-control graph. It also adds bounded primary-source review of two human episodic studies. No new held-out result, recurrent fit, biological dataset or formal compilation is claimed. A fresh prospective evaluation is required after choosing the next learning strategy.

Draft 5 adds a separate frozen active-classifier protocol, twelve learned model-family files containing 96 task trees, training examples, development and held-out results, full query traces, an exact replay, a post-result audit, conditional background intervals and two figures. Earlier neural fits and their adverse results are unchanged. Three active-acquisition references are added with abstract-level access boundaries; no published implementation has been replicated. The abstract is consolidated without removing earlier arguments or numerical results from the body. The new finite witness is explanatory, not a new formal compilation. Independent review, full genealogy, final editorial consolidation and publication packaging remain open.

Draft 6 added the primary mouse-V1 integration/segregation comparison [28] and a prospective test that does not convert its component result into SAN confirmation. This preprint preserves that analysis, all thirteen figures and the later joint-measurement diagnostic, with application receipts and formal scope available in the companion. It does not add participants, animal recordings, a trained integrated recurrent-acquisition system or independent scientific review.

Astra revision 1 preserves that historical final as a separate file and changes no original experiment. Its protocol and algebra addendum, diagnostic source and bounded diagnostic result make the request-cost and additive-parameterization corrections inspectable. The revision receipt gives exact source identities and preservation checks. The learned joint experiment in Section 6.8 remains prospective and unrun. Section 6.9 executes only the separately declared hand-constructed measurement and readout-factorization diagnostic. No fresh held-out learned result, biological validation, Lean recompilation, public upload or publication clearance is claimed.

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